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Review article

Exploring machine learning trends in poverty mapping: A review and meta-analysis

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ABSTRACT

Machine Learning (ML) has rapidly advanced as a transformative tool across numerous fields, offering new avenues for addressing poverty-related challenges. This study provides a comprehensive review and meta-analysis of 215 peer-reviewed articles published on Scopus from 2014 to 2023, underscoring the capacity of ML methods to enhance poverty mapping through satellite data analysis. Our findings highlight the significant role of ML in revealing micro-geographical poverty patterns, enabling more granular and accurate poverty assessments. By aggregating and systematically evaluating findings from the past decade, this meta-analysis uniquely identifies overarching trends and methodological insights in ML-driven poverty mapping, distinguishing itself from previous reviews that primarily synthesize existing literature. The nighttime light index emerged as a robust indicator for poverty estimation, though its predictive power improves significantly when combined with daytime features like land cover and building data. Random Forest consistently demonstrated high interpretability and predictive accuracy as the most widely adopted ML model. Key contributions from regions such as the United States, China, and India illustrate the substantial progress and applicability of ML techniques in poverty mapping. This research seeks to provide policymakers with enhanced analytical tools for nuanced poverty assessment, guiding more effective policy decisions aimed at fostering equitable development on a global scale.

1. Introduction

Poverty mapping is a methodology employed to depict the socio-economic conditions of individuals residing in a specific area, resulting in a comprehensive map that illustrates various properties to identify those most likely to experience poverty. This intricate network can impede progress across multiple domains, including education, socio-economic development, health, and overall quality of life. Poverty often restricts access to quality education, creates barriers to effective healthcare and preventive services, and hampers socio-economic advancement. Furthermore, it adversely affects societal development, fostering social discrimination that leads to isolation and diminished participation in public decision-making processes regarding community development initiatives (Asian Development Bank, 2024). The concept of poverty extends beyond a mere lack of financial resources and encompasses multidimensional aspects that reflect the minimum living standards within a particular society. These standards include the fulfillment of basic needs such as food, shelter, clothing, education, and healthcare. International organizations, including the World Bank and the United Nations, underscore the importance of access to adequate, culturally acceptable, and nutritionally sufficient food, as well as safe and dignified housing that includes sanitation, water, and electricity, all of which significantly influence living standards. Additionally, socio-economic public participation, which encompasses access to education, employment opportunities, and social interactions, is crucial for fostering a sense of empowerment (Fisher, 2008; Yadav, 2021). The United Nations 2030 Agenda for Sustainable Development delineates the principles of equitable and sustainable development for all nations, encapsulated within the 17 Sustainable Development Goals (SDGs). To address these objectives, particularly the eradication of poverty, the agenda acknowledges the ambitious targets aimed at generating income through opportunities, facilitating access to finance and fair wages, and ensuring social protection in healthcare and education. Moreover, it emphasizes the empowerment of individuals to engage in community

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development, thereby influencing broader societal progress (Mirelman et al., 2016; Kolak et al., 2020).

Despite advancements in poverty reduction over recent decades, significant challenges remain. While some regions have seen reductions in poverty levels, an accompanying inequality gap has emerged. Factors such as conflict, pandemics, and climate change disproportionately exacerbate the plight of impoverished populations. The challenge of accurately measuring poverty, particularly through its multidimensional factors persists as a critical issue. Commonly utilized indicators may encompass economic, social, demographic, vulnerability, or participatory metrics depending on context to visualize areas with high poverty concentrations. Developed nations like the United States and several European countries have implemented social safety nets to mitigate poverty rates; however, they continue to face issues of social exclusion. In contrast, narratives from developing countries vary significantly; some have made substantial progress in poverty alleviation while others continue to grapple with persistent challenges. The decade from 1990 to 2000 was pivotal in lifting millions out of poverty; however, many African nations still lag behind (Griggs, 2013; Anon, 2023). The Millennium Development Goals (MDGs) established a global framework for addressing poverty; nevertheless, the COVID-19 pandemic has reversed some gains by pushing step back into poverty by the end of the decade (World and Washington, 1981; McBride and Nichols, 2018). The overarching aim remains ensuring economic growth and sustainable development in an environmentally sustainable manner (Griggs, 2013; Anon, 2020).

Traditional data collection methods such as surveys and censuses present inherent challenges including complexity and time consumption. In contrast, machine learning (ML) offers a more efficient alternative for poverty mapping by addressing issues related to high costs and biases associated with traditional survey methods (Alkire et al., 2015; Anon, 2024). ML can effectively process large datasets from diverse sources across extensive geographical regions, enabling unbiased poverty map creation. However, considerations regarding data quality and privacy must be prioritized when applying ML in this context (Anon, 2023; MDG Gap Task Force Report 2015, 2015). These factors underscore the necessity for systemic improvements to meet the SDGs by 2030. For instance analysis fostered by ML technique shows that the extreme poverty rate decreased from 36.2% in 1990 to 9.2% in 2017, it subsequently rose due to COVID-19 impacts. Although these figures began to decline post-2021, slow progress remains a challenge, particularly in Asian and African nations where achieving SDGs is complicated by various factors. The COVID-19 pandemic is projected to have pushed an additional back-step into poverty by 2020 the first increase in global poverty since the Asian financial crisis (Anon, 2020; United Nations, 1981; Schoch and Lakner, 2020; World Bank, 2022).

The rapid development of ML techniques has spurred extensive research targeting poverty alleviation. Advanced tools, including deep learning models such as Convolutional Neural Networks (CNNs) (Babenko et al., 2017; Okaidat et al., 2021) and Recurrent Neural Networks (RCNNs) (Tang et al., 2016), are being leveraged to analyze diverse data sources like satellite imagery, mobile phone usage, and economic transactions. This expansion underscores the need for a comprehensive review of ML applications in poverty mapping. This paper presents a meta-review and analysis to enhance understanding of machine learning in this domain, focusing on critical aspects of Machine Learning in Poverty (MLP); Patterns in ML Utilization involves identifying trends in the application of ML methods for poverty mapping. Second, Exploring the goals of MLP research and its significant contributions to poverty reduction. Third based on Datasets in MLP which examines the various datasets used in MLP and evaluating their effectiveness and finally Prominent Model that investigate the leading models associated with MLP, providing insights into methodologies and approaches that are gaining traction in combating poverty. This review paper was developed through a systematic literature search and selection process aimed at curating high-quality, relevant research on poverty mapping using machine learning and related development indices. The review began by collecting peer-reviewed articles from Scopus-indexed databases, carefully filtering results with key search terms, such as "poverty mapping", "indices of poverty mapping", "machine learning", and "development", within the publication timeframe from 2014 to 2023. Duplicate studies were identified and excluded through manual inspection to avoid redundancy. All articles were organized into an article bank for preliminary assessment, where titles and abstracts were read to evaluate their relevance. Only papers that met the criteria were retained for in-depth analysis, with non-relevant studies removed to ensure the quality of the selection. This entire paper selection process is illustrated in Fig. 1.

Conducting a meta-analysis holds significant value as it systematically aggregates and synthesizes findings from a diverse body of studies, offering a comprehensive understanding of the applications of machine learning in poverty mapping. This approach uncovers overarching trends and methodological insights and strengthens the reliability of conclusions derived from individual studies, thereby informing policymakers in crafting evidence-based and impactful poverty alleviation strategies. By critically evaluating an extensive body of literature, a meta-analysis elucidates the effectiveness of various machine learning techniques and their contributions to nuanced poverty assessments, addressing the complex and multifaceted nature of poverty across diverse contexts.

2. Progression of data-driven approaches

For decades, the World Bank, UNDP, FAO, and academic institutions have prioritized advancing survey data collection to achieve precise geographic aggregation, thereby enhancing policymaking. Over time, poverty mapping methodologies have undergone substantial evolution. The latest Guidelines for Small Area Estimation (SAE) for Poverty Mapping integrate over three decades of research, presenting a contemporary conceptual framework for analyzing and interpreting poverty in modern contexts. By utilizing survey data, models can be developed to establish the relationship between household welfare metrics such as per capita household expenditure and household attributes, member characteristics, and geographic location. Poverty mapping serves as a critical tool for identifying and targeting the most vulnerable populations. Notably, the Social Security Administration economist Mollie Orshansky defined poverty in the 1960s by tripling the minimal food cost to account for additional household expenses, establishing the poverty threshold. This pivotal definition laid the foundation for progressive advancements in poverty mapping, enabling the identification of increasingly granular determinants (Fisher, 2008). The developmental process is expressed in Fig. 2. This figure encapsulates the evolution of techniques and frameworks that have emerged over the years, highlighting significant milestones in the integration of modeling to enhance the accuracy of poverty assessments globally.

Significant publications from the 1970s, such as the World Bank's World Development Report and UNCTAD's Trade and Development Report, laid a foundational understanding of global development challenges (United Nations, 1981). While the Trade and Development Report analyzed global economic issues from a development perspective, the World Development Report addressed critical development challenges of the 1980s, including poverty and human development. These seminal works underscored the complexity of development and the need for tailored policies to address the multifaceted issues faced by developing nations. The 2013 Nature article, Sustainable Development Goals for People and Planet, by Griggs et al. emphasized the dual priorities of eradicating poverty and safeguarding Earth's life support systems as central to sustainable development (Griggs et al., 2013). The authors identified six overarching objectives crucial for achieving global sustainability and reducing poverty: ensuring food and water security, access to clean energy, healthy ecosystems, prosperous livelihoods, and governance for sustainable communities. Each objective includes

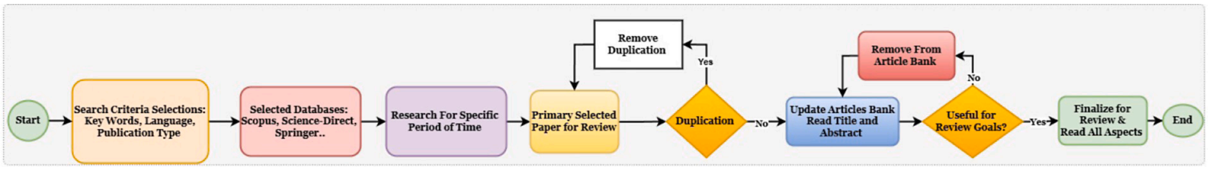

Fig. 1. Systematic literature review workflow for poverty mapping studies. The flowchart illustrates the step-by-step selection process from initial search criteria and database selection through duplication removal and final review on poverty mapping using machine learning and development indices.

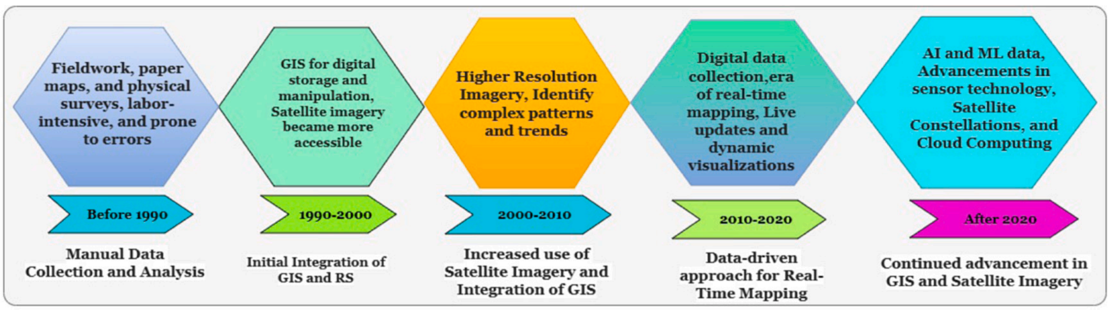

Fig. 2. Evolution of Geospatial Data Capabilities and Applications Over Time indicating the Developmental Process of Poverty Mapping.

specific targets, such as eradicating poverty and hunger, combating diseases like HIV/AIDS, improving maternal and child health, stabilizing the climate, preserving biodiversity, protecting ecosystem services, and promoting the sustainable use of resources (Anon, 2023; Jiang et al., 2019). Capturing regional poverty patterns necessitates the integration of diverse datasets and advanced analytical techniques. Technological advancements have transformed decision making from manual processes to efficient, machine-driven approaches capable of considering multi-integrated factors. High-resolution satellite imagery (Gram-Hansen et al., 2019), point-of-interest (POI) databases (Hurst et al., 2020b; Ledesma et al., 2020), OpenStreetMap (OSM) (Chao et al., 2021), Digital Surface Models (DSMs), and census-derived aggregate data collectively enable a comprehensive understanding of regional environments and the socioeconomic dynamics influencing poverty distribution. To model the complex relationships between poverty and geospatial factors, machine learning algorithms such as Principal Component Analysis (PCA) (Wang et al., 2019), Random Forests (Bakar et al., 2020; Zhao et al., 2019; Tracy et al., 2021; Browne et al., 2021), and Gradient Boosting Trees (Lin Htet et al., 2021; Ayush et al., 2020; Hersh et al., 2020) are extensively employed. By integrating modern analytical methods with a broad spectrum of data sources, researchers can generate more accurate and detailed poverty maps, offering critical insights for targeted interventions and policy formulation. The evolution of poverty mapping and its application history are detailed in the milestone chart, as illustrated in Fig. 3.

In 2017, D.L. Wilson et al. published a case study that examined how ensemble machine learning and forecasting methods could be used to improve rural handpumps maintenance in Western Kenya. They created a predictive model by utilizing sensor data from 42 handpumps manufactured under the Afridev brand. This algorithm was able to forecast pump failures ahead of time and quickly identify existing problems, resulting in an outstanding 99% uptime record. The researchers effectively trained their model by combining sensor observations of pump failures with human-verified data, demonstrating the model's capacity to predict faults and expedite repair procedures. In certain locations, the application of this cutting-edge machine learning solution could greatly improve public health outcomes and raise consumer satisfaction levels (Wilson et al., 2017b). Similarly, Mukelabai et al. released a paper titled "Using Machine Learning to Expound Energy Poverty in the Global South: Understanding and Predicting Access to Cooking with Clean Energy" in 2023. The study explores the use of machine learning to improve understanding and prediction of the accessibility of clean cooking energy in underdeveloped countries. It emphasizes how important it is for financially resilient to have home energy systems and offers workable alternatives for clean cooking. The authors stress the importance of increasing households' financial capacity to purchase electricity or clean cooking fuels to push for a change in approach to address energy poverty. Based on their findings, it is estimated that investment would be necessary to achieve universal access to clean cooking (Mukelabai et al., 2023). Moreover, Csaba Veres and Jennifer Sampson's study "Self-Supervised Learning and the Poverty of the Stimulus" explores the fascinating field of self-supervised learning and clarifies how this method can get over the constraints brought on by the stimulus's lack of resources. Through their exploration of this idea, the authors provide a novel viewpoint on the autonomous learning capabilities of machine learning models from unlabeled data, opening up the window to more effective and efficient learning procedures. Their work opens up new possibilities for improving machine learning algorithms through self-supervision, while also challenging established learning paradigms (Veres and Sampson, 2023).

The issue of artificial intelligence bias in pricing algorithms used by ride-hailing companies was critically examined by A. Pandey and A. Caliskan. Their study revealed discriminatory behavior in these systems, with varying impacts on different demographic groups. An analysis of anonymized data from Chicago highlighted a concerning pattern: neighborhoods with higher proportions of non-white residents, younger populations, lower income levels, and higher educational attainment experienced algorithmic biases leading to increased fares. This research underscores the imperative of ensuring accountability and transparency to mitigate AI's adverse effects in urban contexts (Pandey and Caliskan, 2021). In the context of advancing poverty mapping methodologies, the paper Poverty Mapping in the Age of Machine Learning explores the integration of modern machine learning techniques with remotely sensed data. It critiques the conventional validation process, which typically compares subnational, machine learning-based poverty estimates with direct survey-derived estimates. The study argues that direct estimates may not accurately reflect true poverty rates, potentially leading to misleading model evaluations. Using simulation experiments based on data from the 2015 Mexican Intercensal Survey, the researchers demonstrate that traditional validation methods can obscure the actual performance of models, thereby complicating the selection of optimal approaches across varying scenarios (Corral Rodas et al., 2023). The findings suggest that while machine learning-based poverty estimates can match or exceed traditional methods under certain validation techniques, their

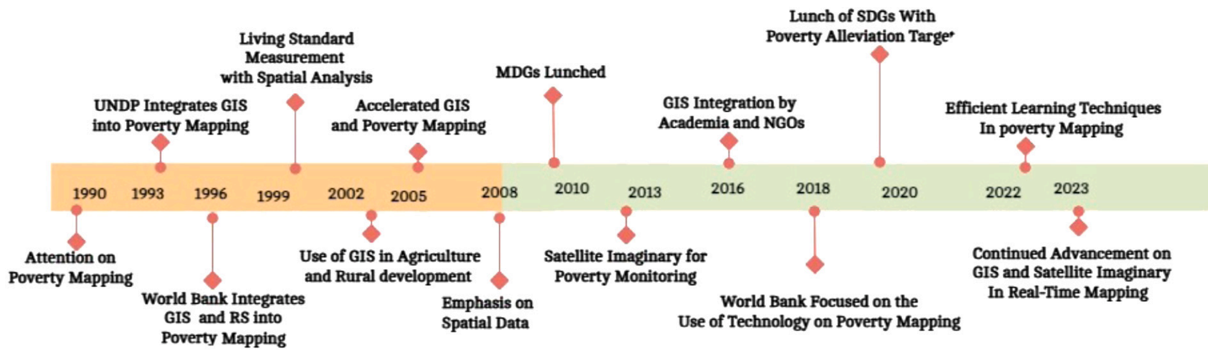

Fig. 3. Milestone representation of the developmental process and utilization of poverty mapping.

performance may falter when measured against "real" poverty rates, particularly when relying on publicly available geographic data.

Grueso in his paper offers a methodological paradigm for causal inference in international development that blends econometric designs with machine learning. The framework is used to investigate the connection between Colombia's multifaceted poverty and violence. A directed acyclic graph (DAG) that forecasts the relationships between the elements of poverty and violence is made using Bayesian networks. Access to water and the condition of walls and flooring are examples of minimum living conditions that strongly predict the health and educational aspects of poverty. A home with an illiterate member has a 0.4% higher chance of experiencing violence, according to research using SLS (Grueso, 2023), conventional poverty mapping techniques. To create more accurate poverty estimates for successful policy interventions it is required to have significant data quality, merging geographical data with higher-quality sources in poverty mapping and proposes (World and Washington, 1981; Corral Rodas et al., 2023; Burke et al., 2021).

3. Methodological framework and data ecosystem

Poverty mapping is a multifaceted process, requiring an integration of data sources, advanced methodologies, and innovative technologies to accurately represent spatial disparities in poverty. The complexities of poverty often manifest as distinct spatial patterns, each influenced by a unique set of variables. To address these variations, effective poverty mapping employs a combination of data sources and methodologies, aiming to uncover the hidden dimensions and distribution of deprivation. By providing granular insights into poverty, this approach informs policymakers and enables the strategic allocation of resources to regions and communities most in need (Griggs et al., 2013; Anon, 2023). A range of data-driven, statistical, econometric, participatory, and hybrid methodologies underpins poverty mapping. These methodologies rely on diverse data sources, including traditional surveys, remote sensing, big data, and community input, to create multidimensional perspectives. Through the use of both direct and proxy measures, such as consumption patterns and basic needs indicators (e.g., sanitation, water, education, housing), poverty mapping enables researchers to estimate welfare at fine spatial resolutions, ensuring sensitivity to local contexts (Zhang et al., 2021b; Martey and Armah, 2021).

3.1. Primary data-driven approaches: Surveys, remote sensing, and big data

Primary data-driven approaches to poverty mapping, including surveys, remote sensing, and big data analytics, are vital for accurately assessing and addressing poverty. While surveys provide essential insights into household-level economic conditions, they often face limitations such as infrequency and high costs (Georganos et al., 2019). In contrast, remote sensing technologies leverage satellite imagery to obtain real-time data on environmental and infrastructural variables related to poverty (Agarwal et al., 2019). Research indicates that combining satellite-derived indicators with mobile phone data can create high-resolution poverty maps that are regularly updated, enhancing decision-making for targeted interventions. Furthermore, the use of big data analytics and machine learning algorithms enables the processing of extensive datasets to identify complex patterns of poverty across different regions. This multifaceted approach improves the granularity of poverty assessments and facilitates timely responses to emerging vulnerabilities, thereby supporting efforts to eradicate poverty in alignment with global development objectives (González-Vidal et al., 2017).

3.1.1. Traditional surveys and censuses: Foundation of poverty mapping

Traditional survey and census data form the backbone of poverty mapping by offering extensive, representative information on socioeconomic variables. These instruments typically assess poverty through metrics such as income, healthcare, education, and access to sanitation, thereby providing a baseline for spatially resolved poverty estimates. However, limitations arise due to data collection frequency and potential biases, necessitating supplementary methods to enhance the robustness and resolution of poverty estimates (Pandey et al., 2018). Small Area Estimation serves as a crucial tool in bridging these gaps by refining spatial detail through statistical modeling, allowing poverty rates to be estimated at granular scales. This approach provides a nuanced picture of poverty distribution across various geographies, equipping policymakers with precise, actionable insights for targeted intervention (Alsharkawi et al., 2021; Crystal and Irvine, 2019; Jarry et al., 2021). When combined, traditional surveys, censuses, and SAE offer comprehensive data points, aiding the creation of informed poverty reduction strategies (Newhouse, 2024).

3.1.2. Remote sensing: Satellite-based insights for poverty indicators

Remote sensing introduces an innovative, high-resolution layer to poverty mapping by providing detailed spatial information over large, often inaccessible regions. By analyzing satellite imagery, researchers can infer poverty indicators through proxies such as housing materials, road networks, and infrastructure presence. These visual data points reveal complex patterns of settlement quality, resource accessibility, and infrastructure disparity, all of which contribute to the spatial understanding of poverty (Pandey et al., 2018; Wang et al., 2021a; Dharmasena et al., 2016). Through remote sensing, poverty mapping extends beyond static measurements of income or consumption, capturing dynamic environmental and infrastructural conditions that influence poverty levels. This bird's-eye perspective is invaluable for identifying regions with infrastructural deficits, often correlated with higher poverty rates. However, the limitations of remote sensing, such as resolution constraints and the need for updated datasets, necessitate a balanced integration with other methodologies to achieve a holistic perspective on poverty dynamics (Gram-Hansen et al., 2019; Steele et al., 2017).

3.1.3. Big data: Harnessing digital footprints

Big data offers unprecedented insights into the complexities of poverty by analyzing digital interactions and behaviors, such as financial transactions, social media activity, and mobile phone usage. This data provides a unique window into the lifestyle, consumption patterns, and vulnerabilities of economically disadvantaged populations, revealing subtle trends that conventional data might overlook (Bakar et al., 2020; Zhao et al., 2019; Tracy et al., 2021; Browne et al., 2021). Although big data holds significant promise, challenges related to ethics, data quality, and interpretation remain paramount. Issues such as data bias, privacy concerns, and the differentiation between correlation and causation must be addressed to ensure that big data is used responsibly. When managed correctly, big data enriches poverty mapping by delivering real-time insights, helping to craft targeted poverty alleviation strategies that reflect the evolving nature of deprivation (Palacios et al., 2021).

3.2. Statistical and econometric techniques

Statistical and econometric approaches are pivotal in creating a multidimensional perspective on poverty, moving beyond traditional income-based assessments. Multidimensional Poverty Indices (MPIs), such as the Alkire-Foster MPI and Human Development Index (HDI), assess poverty across variables like health, education, and living standards, offering a comprehensive view of deprivation (Oldiges; Beja, 2021). Despite their depth, these indices face challenges in defining relevant deprivations and ensuring comparability across regions. Spatial econometric models, such as Spatial Autoregressive (SAR) and Spatial Error Correction (SEC) models, provide valuable insights by accounting for geographic influences on poverty. By incorporating variables like infrastructure accessibility and proximity to markets, these models uncover complex, spatially dependent poverty patterns. Such models require careful selection and high-quality data to enhance understanding of poverty's spatial dimensions (Kolak et al., 2020; Thongdara et al., 2012). Machine learning algorithms offer transformative potential in poverty mapping, allowing for intricate pattern recognition and prediction. Techniques such as decision trees and random forests capture complex relationships in data, facilitating the accurate classification and prediction of poverty status based on various socioeconomic indicators (Xu et al., 2021b; Ayush et al., 2020; Hurst et al., 2020a). However, issues such as algorithmic bias and interpretability require careful consideration to ensure the responsible application of machine learning in poverty mapping (Bakar et al., 2020; Zhao et al., 2019; Tracy et al., 2021; Browne et al., 2021).

3.3. Participatory methods: Integrating local knowledge and community perspectives

Community mapping and Participatory Poverty Assessments (PPAs) empower communities by integrating local voices and experiences into the analysis of spatial challenges and poverty dynamics. These tools enable communities to visually articulate their lived experiences, providing a robust foundation for analysis, advocacy, and action. By highlighting critical issues such as resource scarcity, high-risk areas, and socio-economic vulnerabilities, these participatory approaches foster a sense of ownership and agency among participants. Community members engage directly in the process, traversing their neighborhoods to annotate key features through manual sketches or digital tools. The resulting maps not only reflect nuanced social and spatial realities but also serve as powerful instruments for advocacy, informing local policies and driving targeted interventions (Anon, 2023; MDG Gap Task Force Report 2015, 2015; Toerien, 2023; World Bank, 2022; Robb, 2002). Through their community-driven nature, these approaches generate inclusive and sustainable outcomes by aligning solutions with local priorities and spatial dynamics. The collective act of mapping cultivates a shared understanding, enabling communities to co-create strategies that resonate with their lived realities. Active participation from all segments of the community, particularly marginalized groups, ensures a comprehensive representation of needs, which is essential for crafting impactful and equitable solutions (Martey and Armah, 2021; Shin et al., 2018).

However, for these processes to achieve their full potential, several challenges must be addressed. Ensuring inclusivity and overcoming technological barriers such as the need for training and access to mapping tools—are critical for success. Furthermore, the sustainability of these initiatives depends on their ability to inspire tangible actions and long-term support. Without sustained engagement, the transformative potential of community mapping and PPAs may remain unrealized. By combining spatial awareness with participatory decision-making, these tools offer a pathway towards inclusive development and meaningful community empowerment (Toerien, 2023).

3.4. Hybrid approaches: Integrating multisource data for holistic poverty analysis

To achieve a comprehensive understanding of poverty, it is essential to adopt a multifaceted perspective rather than relying on a single focal point. Hybrid approaches to poverty mapping exemplify this concept by integrating diverse data sources and methodologies to construct a more holistic and realistic representation. For instance, satellite imagery and survey data form the foundation by capturing income levels and access to services, while remote sensing data elucidates geographical patterns. Government records contribute insights into vulnerabilities, and mobile phone usage data reveals real-time mobility patterns. This combination not only addresses the "what" of poverty but also uncovers the "where" and "why", thereby offering a nuanced narrative of poverty's dimensions and drivers (Aiken et al., 2020; Steele et al., 2017).

Triangulation techniques further enrich this understanding by combining quantitative data with the qualitative depth of PPAs and social mapping. Quantitative survey results are contextualized with community voices, highlighting underlying challenges, concerns, and lived experiences. Mapping workshops transform statistical data into dynamic narratives, revealing regional perspectives and spatial trends. This integrative approach ensures that proposed solutions align with the realities of those affected, mitigates bias, and validates findings. Successful integration of diverse data sources requires meticulous planning and technical expertise, supported by a range of skills and resources. Ethical considerations, particularly regarding data privacy and informed consent, are paramount. Despite these challenges, the advantages are transformative: a deeper understanding of poverty, targeted interventions, and ultimately, a more promising future for marginalized populations. Hybrid techniques thus emerge as a cornerstone for devising solutions that not only address poverty but also empower communities, offering actionable insights and fostering meaningful change (Corral Rodas et al., 2023; Toerien, 2023).

3.5. Advances in satellite imaging and machine learning for mapping

The field of poverty mapping has experienced a paradigm shift with the integration of satellite imaging, offering an unprecedented abundance of precise and analytically rich data. Comprehensive datasets, including WorldPop (Depsky et al., 2022; Gaughan et al., 2016), LandScan (Njuguna and McSharry, 2017; Omitaomu et al., 2021), and the Global Human Settlement Layer (Puttanapong et al., 2020), provide detailed insights into population distribution and settlement patterns. These datasets are instrumental in identifying areas with potentially high poverty rates. Additionally, nighttime light data from sources such as VIIRS (Ivan et al., 2020) and LAN serves as a proxy for economic activity, with regions exhibiting low light emissions often associated with higher levels of poverty. High-resolution imagery from platforms such as Stanford (Yadav, 2021; Ayush et al., 2021) and Kaggle (Yeh et al., 2020) further enriches this analysis by incorporating data on human activity, infrastructure, and land use. When processed through advanced computational techniques, these data sources fuel machine learning algorithms, including Random Forests (Bakar et al., 2020; Zhao et al., 2019; Tracy et al., 2021; Browne et al., 2021) and Neural Networks (Tracy et al., 2021; Khaefi et al., 2019; Subash et al., 2018; Kherwa et al., 2021; Monlezun et al., 2021), enabling precise poverty predictions. Additional datasets, such as those available on FigShare (Jiang et al., 2021) and vegetation indices derived from MODIS (Hersh et al., 2020; Wang et al., 2021a; Sedda et al., 2015), provide complementary insights that enhance the overall analytical framework. Table 1 presents a comparative analysis of these major data sources, emphasizing their properties and applications, while Appendices A and C detail the sources, associated studies, methodological applications, and accuracy metrics.

Achieving a comprehensive understanding of poverty requires the integration of multiple datasets to construct a multidimensional representation of its underlying dynamics. This approach enables the identification of complex and interconnected processes contributing to poverty. However, this analytical capability must be employed with caution and responsibility. Ethical considerations, such as data privacy and informed consent, must guide each step to ensure that poverty mapping initiatives empower and uplift affected communities rather than perpetuating disparities. The synergy between satellite imaging and machine learning offers transformative potential for detailed and accurate poverty mapping, provided that the principles of inclusivity, accuracy, and ethical conduct are upheld. By adhering to these standards, technological advancements can be effectively leveraged to address global poverty challenges and foster sustainable solutions (Okaidat et al., 2021; Kherwa et al., 2021; Bansal, 2020).

4. Leveraging ML and AI techniques for poverty analysis and alleviation

Advancements in machine learning (ML) are reshaping poverty mapping by offering a cost-effective, granular alternative to traditional survey-based methods, which often struggle with limited coverage and high expenses (Mirelman et al., 2016; Martey and Armah, 2021; Puttanapong et al., 2020; Pangestu et al., 2020). ML's powerful analytical capabilities allow for the integration of diverse data sources, including satellite imagery, mobile phone records, and social media activity, significantly expanding the precision and scope of poverty predictions. Satellite data provides high-resolution insights into land use, infrastructure, and housing characteristics factors closely associated with socioeconomic conditions. Patterns in land use and infrastructure accessibility reveal levels of economic development, while indicators like building density and housing quality further enhance poverty assessments (Ledesma et al., 2020; Pandey et al., 2018; Kherwa et al., 2021; Ni et al., 2020; Buddaraju et al., 2021; Jean et al., 2016; Wolk et al., 2021; Brahma and Mukherjee, 2021; Tingzon et al., 2019; Zhao et al., 2019; Wang et al., 2021b; Georganos et al., 2019; Hipp et al., 2017; Hu et al., 2020; Babenko et al., 2017; Engstrom et al., 2019b; Steele et al., 2017; Engstrom et al., 2017a). Mobile phone data, encompassing call records and app usage, offers critical insights into population density, movement patterns, and economic behaviors, enabling further refinement of poverty estimates. Social media data provides yet another layer of understanding by revealing public discourse and user behaviors that may indicate regions with limited resources or economic distress (Aiken et al., 2020; Crystal and Irvine, 2019; Hernandez et al., 2017; Khaefi et al., 2019; Khan and Blumenstock, 2019; Oughton and Mathur, 2021). By synthesizing these diverse data streams, ML models can create highly detailed poverty maps, filling gaps in survey data and revealing subtle correlations between poverty indicators and demographic factors. These detailed poverty maps enable precise resource allocation, allowing aid to reach the most vulnerable communities efficiently. Furthermore, by analyzing demographic and socioeconomic data, ML algorithms support the development of early warning systems that predict poverty surges, triggered by economic downturns or disasters, enabling timely intervention to mitigate impacts (Mirelman et al., 2016; Pandey and Caliskan, 2021).

4.1. Supervised learning

To predict poverty indicators and produce accurate poverty maps, supervised learning techniques have become an integral tool in poverty analysis. These methods leverage diverse data sources, such as geographical features, mobile phone records, and satellite imagery, to identify poverty hotspots and quantify the extent of deprivation in specific regions. Among these techniques, regression models, including Support Vector Regression (SVR) and Random Forest Regression, have demonstrated significant effectiveness in correlating data sources with poverty indicators such as income levels and housing quality (Brahma and Mukherjee, 2021; Irvin et al., 2020; Mukherjee et al., 2021; Li and Yao, 2018; Zegarra and MacHicao, 2021). For instance, SVR is capable of predicting poverty rates by analyzing specific housing attributes extracted from satellite imagery, such as roof types or the presence of power lines (Li et al., 2021). Similarly, Random Forest Regression utilizes an ensemble of decision trees (Bakar et al., 2020; Zhao et al., 2019; Tracy et al., 2021; Browne et al., 2021) to generate predictions, excelling at capturing intricate relationships within the data, even when the dataset size is limited (Ramesh et al., 2021).

The Random Forest (RF) model is a robust approach that combines bagging ensembles with random subspace methods. Its advantages include reduced sensitivity to multicollinearity and the capability to handle missing or imbalanced data effectively. The methodology begins by creating bootstrap-based resampled subsets of the original dataset, which are then employed to construct multiple decision trees. The outputs of these decision trees are aggregated to produce the final predictions or classifications. In addition to its predictive and classification capabilities, the RF model is effective at mitigating overfitting and improving prediction accuracy through averaging (Zhao et al., 2019; Tracy et al., 2021).

Support Vector Machines (SVMs) (Crystal and Irvine, 2019; Hernandez et al., 2017; Khaefi et al., 2019; Saini et al., 2020; Li et al., 2019) represent an advanced machine learning approach well-suited for small sample sizes, nonlinear relationships, and high-dimensional datasets. SVMs have demonstrated robust predictive performance (Yin et al., 2021), relying on an optimal separating hyperplane to partition data into distinct classes. By maximizing the margin and ensuring the largest possible distance between data points on either side of the hyperplane, SVMs reduce the upper bound of generalization error. Notably, the model's complexity is independent of the number of features in the training dataset. The primary objective of SVMs is to identify a hyperplane equation that effectively divides the data, ensuring that all points of the same class remain on one side (Bansal, 2020; Li et al., 2021; Xu et al., 2021a; Puttanapong et al., 2021).

Naive Bayes (NB) (Shen et al., 2021; Wolk et al., 2021; Xu et al., 2021a; Alsaqabi et al., 2019) is a machine learning algorithm grounded in Bayes' theorem, assuming conditional independence among features. The algorithm first learns the joint probability distribution of inputs and outputs from a training dataset based on this independence assumption. Subsequently, it predicts outputs by maximizing the posterior probability derived from the learned model. Known for its computational efficiency, the NB algorithm performs exceptionally well on large datasets and is among the fastest classification techniques available. Its assumption of feature independence simplifies computation while delivering robust classification outcomes (Alsaqabi et al., 2019).

Table 1
Comprehensive overview of major datasets for effective utilization in poverty mapping, estimation, and analysis.

Dataset name	Class activity	Size (approx.)	Resolution	Properties	Application area	Ref
WorldPop	Demographics	1.5 TB	Various (100 m to 1 Km)	High spatial and temporal resolution with grid Population	Population Density Estimation and Settlement distribution	Depsky et al. (2022), Gaughan et al. (2016)
Global Human Settlement Layer (GHSL)	Settlements	1.5 GB	250 m	Location-based human settlements	Prone to high poverty concentration and urban/rural classification	Puttanapong et al. (2020)
LandScan	Demographics	800GB	30 m	High resolution with grid population	Spatial Analysis and Identification of small pockets	Njuguna and McSharry (2017), Omitaomu et al. (2021)
Light at Night(LAN)	Economic Activity	1TB	Various range from 250 m to 1 Km	Night light emissions	Poverty mapping based on estimating economic activity	Ivan et al. (2020), Ni et al. (2020), Saini et al. (2020)
Kaggle Dataset	ML for diverse field	100 GB	Varies 224 × 224,512 × 512 or 1024 × 1024 pixels	Satellite images mapping poverty labels	Training ML models for poverty mapping	Yeh et al. (2020)
Stanford University Study	ML for diverse field	100 GB	Varies 224 × 224,512 × 512 or 1024 × 1024 pixels	Satellite image and demographic data	Testing and validating poverty prediction models	Yadav (2021), Ayush et al. (2021)
FigShare Dataset	Socio-economic learning	varies depending on dataset	Various range may be Spatial or Non-Spatial	varies depending on dataset	Poverty mapping based on household surveys and infrastructure data	Jiang et al. (2021)
MODIS Vegetation Indices	Environment factors	1TB	Varies 250 m to 1 Km	Vegetation and greenness data	Poverty mapping based on Land Use, Agricultural Productivity	Hersh et al. (2020), Wang et al. (2021a), Sedda et al. (2015)
VIIRS Nighttime Lights	Economic Activity	1TB	500 m	Nighttime light data (VIIRS sensor)	Poverty mapping with a higher temporal resolution	Ivan et al. (2020)

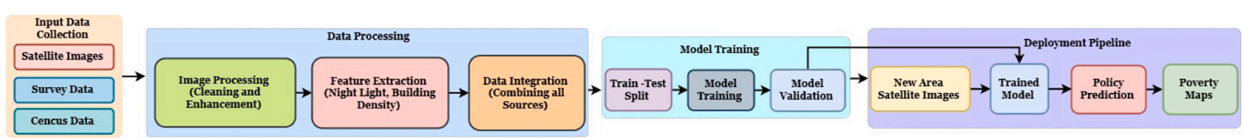

Fig. 4. Supervised Machine Learning Pipeline for Poverty Mapping: A label-based approach that learns from known poverty data (survey and census) combined with satellite imagery to train a predictive model. The pipeline includes data processing, model training with ground truth labels, and deployment for predicting poverty levels in new areas.

Additional supervised learning methods have been employed in poverty mapping to enhance predictive accuracy. Decision trees are frequently used to estimate poverty levels by segmenting data based on relevant poverty indicators. K-Nearest Neighbors (KNN) (Wang et al., 2021a; Guo et al., 2021; Chen and Lian, 2021) is another effective technique that groups regions based on feature similarity, facilitating straightforward yet efficient classification. Logistic regression (Wolk et al., 2021; Wang, 2021b) is commonly applied to predict the probability of a region falling below the poverty line, considering various influencing factors. Ensemble learning techniques, such as Gradient Boosting (Buddaraju et al., 2021; Mukherjee et al., 2021; Chakrabarty and Biswas, 2018) and AdaBoost (Nica-Avram et al., 2021; Saini et al., 2020; Guo et al., 2021), combine multiple models to improve prediction performance. Furthermore, neural networks, including feedforward and recurrent architectures, are increasingly utilized in poverty mapping to capture complex relationships between input features and poverty indicators, further advancing the field (Khaefi et al., 2019; Monlezun et al., 2021; Buddaraju et al., 2021). Fig. 4 shows the stages of supervised learning to predict the poverty.

4.2. Unsupervised learning

Unsupervised learning techniques play a pivotal role in uncovering latent patterns and extracting meaningful insights from datasets that lack explicit labels. These techniques are particularly valuable in poverty analysis, where labeled data is often scarce. Clustering algorithms, such as k-means clustering (Liu et al., 2021b; Rincón, 2019; Deng et al., 2017; Garwood and Dhobale, 2021; Wahidah and Utari, 2023), are widely employed to group similar data points based on shared features. For instance, satellite imagery can be clustered based on infrastructure characteristics and land use patterns, enabling the identification of potential poverty hotspots even in the absence of explicitly labeled data. Such insights can inform targeted policy interventions and resource allocation for poverty alleviation. Principal Component Analysis (PCA) (Abson et al., 2012) is one of the most widely utilized dimensionality reduction techniques for transforming high-dimensional data into a lower-dimensional space while preserving critical information. By simplifying satellite imagery data, PCA facilitates the extraction of key features associated with poverty estimation, such as building density, vegetation cover, and road networks. This reduction in complexity allows researchers to identify the most significant factors influencing poverty levels in various regions. Moreover, PCA supports the visualization of complex datasets, aiding in the interpretation of socioeconomic disparities across geographic areas.

Beyond clustering and dimensionality reduction, unsupervised learning techniques significantly enhance poverty mapping by revealing hidden patterns and relationships that may not be immediately apparent in the raw data. For example, these methods can identify

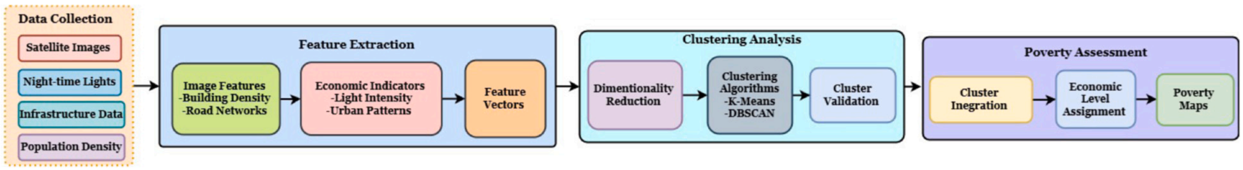

Fig. 5. Unsupervised Machine Learning Pipeline for Poverty Mapping: A data-driven approach that automatically discovers patterns and clusters regions based on satellite imagery, nighttime lights, infrastructure, and population density without requiring labeled training data. The process involves feature extraction, clustering analysis, and poverty level assessment through cluster interpretation.

correlations between urbanization, land use changes, and poverty indicators, providing valuable insights for policymakers. By enabling a more nuanced understanding of socioeconomic dynamics, unsupervised learning contributes to the development of evidence-based strategies for addressing poverty. Consequently, these techniques play a crucial role in advancing the field of poverty analysis, offering innovative solutions for tackling complex challenges in policy-making and development planning. Fig. 5 illustrates the flow of unsupervised learning in a block diagram, providing a visual representation of their respective methodologies in context of poverty mapping. This diagram underscores the complementary nature of these approaches in uncovering actionable insights and guiding effective interventions.

4.3. Deep learning

Deep learning algorithms provide advanced methodologies for deriving insights from complex data sources. In the domain of poverty mapping, CNNs have emerged as a pivotal tool, renowned for their proficiency in image detection and analysis (Babenko et al., 2017; Okaidat et al., 2021; Khan and Blumenstock, 2019). For instance, CNNs enable the extraction of latent features from satellite imagery, such as the dimensions of buildings and the configuration of road networks, which serve as significant proxies for assessing poverty levels across diverse regions. Similarly, Recurrent Neural Networks demonstrate exceptional performance in handling sequential data, making them a valuable addition to poverty mapping frameworks (Tang et al., 2016; Rajagopal et al., 2021). RNNs have been employed to analyze mobile phone call records for forecasting poverty risk (Aiken et al., 2020; Hernandez et al., 2017; Khaefi et al., 2019; Khan and Blumenstock, 2019; Oughton and Mathur, 2021). By examining mobility patterns derived from call data, RNNs can identify regions characterized by minimal economic activity, thereby pinpointing areas potentially associated with elevated poverty levels. The synergistic application of CNNs and RNNs enhances the analytical capacity of poverty mapping by uncovering intricate patterns and inter dependencies from heterogeneous data sources. Through these sophisticated approaches, researchers and policymakers are equipped to achieve a more nuanced understanding of poverty dynamics, enabling the development of targeted interventions that effectively address socio-economic challenges.

Mapping poverty through deep learning techniques involves a systematic, multi-step methodology. The process commences with the acquisition of high-resolution datasets, predominantly satellite imagery, which provide detailed information on infrastructure, land use, and human activities. To ensure precise geographic population insights, key datasets such as LandScan (Njuguna and McSharry, 2017; Omitaomu et al., 2021) and WorldPop (Depsky et al., 2022; Gaughan et al., 2016) are integrated. These datasets offer accurate spatial population distributions, which are essential for contextualizing poverty-related analyses. The identification of impoverished settlements is further refined through resources like the Global Human Settlement Layer (GHSL) (Puttanapong et al., 2021; European Commission, 2024), which delineates economically disadvantaged regions. Complementary to these datasets, nighttime light data, derived from sources such as VIIRS (Ivan et al., 2020) and LAN, serve as significant economic indicators, with lower levels of light emissions frequently correlating with heightened poverty. Following the collection of these diverse datasets, they are input into CNNs to facilitate advanced feature extraction. CNNs enable the precise detection of spatial patterns related to poverty by analyzing indicators embedded within high-resolution imagery. This approach ensures the accurate mapping of economic conditions by leveraging a combination of rich, heterogeneous datasets. The integration of these datasets with CNNs underscores the transformative potential of deep learning in poverty mapping, enabling comprehensive and data-intensive analyses, as illustrated in Fig. 6.

In the context of machine learning, Classification (Peng et al., 2019; Mayani and Itagi, 2021; Sianes et al., 2014; Kshirsagar et al., 2017) represents a fundamental supervised learning technique wherein data items are categorized based on predefined criteria derived from their attributes. This approach is indispensable in applications such as sentiment analysis and image recognition, where accurate and consistent categorization is paramount. In contrast, Clustering (Liu et al., 2021b; Deng et al., 2017; Mahmood et al., 2021), an unsupervised learning technique, identifies latent patterns within datasets and is frequently employed for tasks such as anomaly detection and consumer segmentation. Similarly, Regression (Zhao et al., 2019; Liu et al., 2021a), another supervised learning methodology, models quantitative relationships between input features and numerical outcomes, enabling predictive tasks like stock price forecasting and sales estimation. These three methodologies collectively form the foundational pillars of machine learning, enabling diverse applications in predictive analytics and pattern recognition. A detailed comparison of classification, regression, and clustering algorithms, along with examples of their deployment across various studies, is summarized in Table 2.

However, the practical implementation of these techniques requires a meticulous approach, particularly with respect to ethical considerations and technological constraints. Ensuring data quality and representativeness is critical to mitigate biases that could lead to inequitable outcomes or erroneous conclusions. In the case of poverty mapping, ethical priorities must be at the forefront, avoiding discriminatory applications while rigorously protecting privacy and ensuring data security. Moreover, the inherent complexity and opacity of certain ML models pose significant challenges, particularly in terms of interpretability, which is crucial for validating predictions and fostering trust. To address these challenges, a strong commitment to fairness, transparency, and interpretability is essential in the evaluation and deployment of ML models. Furthermore, the development and maintenance of ML systems demand substantial investments in expertise and infrastructure, highlighting the necessity for sustainable, long-term strategies to ensure the effective and ethical utilization of these technologies.

5. Assessment metrics for ML in poverty mapping

A range of rigorous evaluation metrics is essential for accurately assessing the performance of machine learning pipelines in poverty mapping. This section outlines the most widely used metrics in machine learning to map poverty divided by the task.

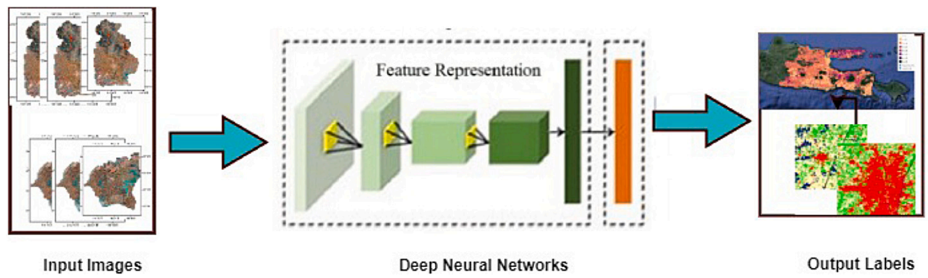

Fig. 6. Block diagram illustrating the workflow of a deep learning model for image analysis, highlighting the transition from input images through feature extraction in deep neural networks to the generation of output labels for Poverty Mapping.

Table 2
Comprehensive overview of prominent machine learning models and their applications in the field of poverty mapping.

MLModels	Types	Uses	Ref
	Naive Bayes	Effective categorization with limited data.	Shen et al. (2021), Wolk et al. (2021), Xu et al. (2021a), Alsaqabi et al. (2019)
	Gradient Boosting	Reduces mistakes to increase accuracy.	Buddaraju et al. (2021), Mukherjee et al. (2021), Chakrabarty and Biswas (2018)
	KNN (K-Nearest Neighbors)	Based on comparable neighbors, KNN (K-Nearest Neighbors) classifies.	Wang et al. (2021a), Guo et al. (2021), Chen and Lian (2021)
	Logistic Regression	Provides comprehensible insights into binary categorization	Wolk et al. (2021), Wang (2021b)
	XGBoost	Accurate and quick handling of intricate relationship data	Xu et al. (2021a), Hossain et al. (2019), Hong and Park (2021), Longa et al. (2021)
Classification	Artificial Neural Network (ANN)	potent but data-hungry mapping technology.	Subash et al. (2018), Prasetyo et al. (2021), Yin et al. (2021), Othman et al. (2020), Viloria et al. (2021)
	Neural Network	potent but data-hungry mapping technology.	Tracy et al. (2021), Kherwa et al. (2021), Subash et al. (2018), Khaefi et al. (2019), Monlezun et al. (2021)
	Adaboost	Increases precision by concentrating on incorrect classifications.	Nica-Avram et al. (2021), Saini et al. (2020), Guo et al. (2021)
	Convolutional Neural Network (CNN)	Excellent for socioeconomic analysis based on images.	Babenko et al. (2017), Okaidat et al. (2021), Khan and Blumenstock (2019)
	Mobilenetv2	Mobile device mapping that is efficient.	Kim et al. (2021)
	ResNet50	Capable of managing intricate features for mapping.	Okaidat et al. (2021), Wolk et al. (2021), Wu and Tan (2019), Kondmann and Zhu (2020), Piaggesi et al. (2019)
	Random Forest	Using an ensemble of decision trees, forecast continuous poverty indicators with higher accuracy.	Bakar et al. (2020), Zhao et al. (2019), Tracy et al. (2021), Browne et al. (2021)
	Support Vector Machine (SVM)	Use hyperplanes to classify.	Bansal (2020), Xu et al. (2021a), Li et al. (2021), Puttanapong et al. (2021)
	Decision Tree	Determine and categorize the causes of poverty.	Ayush et al. (2020), Xu et al. (2021b), Hurst et al. (2020a)
	K-Nearest Neighbors(KNN)	Use neighbors to determine level.	Wang et al. (2021a), Guo et al. (2021), Chen and Lian (2021)
	Logistic Regression	Model probability based on predictors using logistic regression.	Wolk et al. (2021), Wang (2021b)
Regression	XGBoost	Enhance trees for precise recognition.	Xu et al. (2021a), Hossain et al. (2019), Hong and Park (2021), Longa et al. (2021)
	Recurrent Neural Network (RNN)	temporal pattern of crops across image time	Tang et al. (2016), Rajagopal et al. (2021),
	Lasso	effective mapping and prediction, and choose important poverty indicators.	Wang et al. (2019), Engstrom et al. (2019b), Kambuya (2020), Njuguna and McSharry (2017), Verme (2020), Engstrom et al. (2017a), Irvin et al. (2020), Agarwal et al. (2019)
	LightGBM	Boosted decision trees are an efficient way to forecast continuous poverty indicators,	Okaidat et al. (2021), Ledesma et al. (2020), Ramesh et al. (2021)
	K-Means	Group areas with comparable socioeconomic circumstances using K-Means.	Liu et al. (2021b), Rincón (2019), Deng et al. (2017), Garwood and Dhobale (2021), Wahidah and Utari (2023)
	DBSCAN	Locate geographical clusters with comparable circumstances	Omitaomu et al. (2021)
Clustering	Hierarchical Clustering	Discover connections between areas with comparable socioeconomic characteristics.	Rachmawati and Pusponegoro (2018), Liu et al. (2021b)
	Gaussian Mixture Model (GMM)	Spot areas with a range of traits and poverty levels.	Wahidah and Utari (2023)
	Mean Shift	Find areas where poverty is concentrated.	Liu et al. (2021b), Mahmood et al. (2021), Peng et al. (2019)
	Spectral Clustering	find groups that share comparable poverty characteristics.	Gram-Hansen et al. (2019), Yeh et al. (2020)

5.1. Performance comparison

To rigorously assess model configurations and establish meaningful benchmarks, we focus on a targeted subset of performance metrics that enable reliable comparisons across models. Given the challenges of cross-paper comparisons due to variations in datasets and evaluation criteria, we prioritize core metrics: accuracy and F1 score for classification models, alongside R^2 and adjusted R^2 for regression models. Accuracy offers an overall performance metric for classification tasks but it can be insufficient in imbalanced datasets. The F1 score complements accuracy by taking into account precision and recall, making it especially useful when the costs of misclassification are not evenly distributed. For regression models, R^2 indicates how well the independent variables explain the variance in the dependent variable, while Adjusted R^2 takes into account the number of predictors, which penalizes overfitting. Adjusted R^2 is therefore more robust in comparisons between models with varying numbers of features, providing a clearer measure of model generalization. This approach emphasizes metrics that balance interpretability and robustness, ensuring consistent and meaningful evaluation of model performance while accounting for dataset-specific nuances and feature variability.

5.2. Metrics for classification

Three primary metrics are frequently used to assess the results of classification tasks in multilayer analyzers: accuracy, precision, and F1 score. These metrics collectively provide a comprehensive evaluation of model performance, capturing overall correctness, the quality of positive predictions, and the balance between precision and recall.

5.2.1. Accuracy

Accuracy refers to the percentage of cases that the classifier correctly classified, serving as a measure of overall model performance. It is calculated as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where: TP is the number of true positives (correctly classified positive cases) TN is the number of true negatives (correctly classified negative cases) FP is the number of false positives (incorrectly classified positive cases) FN is the number of false negatives (incorrectly classified negative cases).

5.2.2. Precision

Precision measures the likelihood that, given a positive prediction, the classifier has correctly identified a true positive It evaluates the accuracy of positive predictions, making it particularly important in scenarios where false positives carry significant consequences. It is calculated as:

$$Precision = \frac{TP}{TP + FP} \tag{2}$$

where:

TP is the number of true positives FP is the number of false positives.

5.2.3. F1 score

The F1 Score is a metric that combines precision and recall (sensitivity) into a single value, offering a balanced assessment of a model's performance, particularly in cases of imbalanced datasets. It is calculated as the harmonic mean of precision and recall

$$F1Score = \frac{2 \cdot (\text{Precision} \cdot \text{Recall})}{\text{Precision} + \text{Recall}} \tag{3}$$

where Precision and Recall are defined as above and this metric is especially useful when the cost of false positives and false negatives needs to be carefully balanced.

5.3. Metrics of regression

In MLPs the most commonly used metrics for assessing regression tasks include the coefficient of determination (R^2), mean squared error (MSE), and root mean squared error (RMSE). The R^2 metric evaluates the quality of the model's fit by quantifying the proportion of variance in the dependent variable explained by the independent variables. Meanwhile, MSE measures the average squared difference between predicted and actual values, and RMSE provides an interpretable measure of the typical magnitude of prediction errors, expressed in the same units as the target variable. Together, these metrics offer a comprehensive evaluation of model performance in regression tasks.

R^2= coefficient of determination is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n} (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n} (y_i - \bar{y})^2} \tag{4}$$

where $\hat{y}_i$ is the predicted value, y_i is the actual value, and $\bar{y}$ is the mean of the actual values.

The Mean Squared Error (MSE) is calculated as:

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \tag{5}$$

where n is the number of observations, y_i is the actual value, and $\hat{y}_i$ is the predicted value.

The Root Mean Squared Error (RMSE) is calculated as:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{n}} \tag{6}$$

There is cause for trust in the validity and reliability of the recommended evaluation measures due to their widespread acceptance and documented effectiveness in the literature. This makes it possible for ML to be consistently and fairly assessed in applications related to poverty mapping thanks to these measures. The values of R^2, MSE, RMSE, F1 Score are tabulated in Appendix C with the dataset used by the respective authors as their findings.

6. Meta-analysis

6.1. Data collection and strategy

The field of poverty mapping has undergone significant advancements with the integration of machine learning techniques, offering novel approaches to enhancing its precision and depth. This meta-analysis aims to critically examine the current state-of-the-art in applying machine learning to poverty mapping by systematically reviewing a carefully curated selection of academic articles and reports. The primary objectives are to elucidate the methodological frameworks, assess the performance metrics, and evaluate the data sources employed in these studies. The analysis adopts a dual approach: a qualitative assessment to identify the strengths and limitations of individual studies and a quantitative meta-analysis to evaluate the comparative effectiveness of different machine learning algorithms in poverty mapping applications. By synthesizing insights from these dimensions, the review and meta analysis seeks to uncover procedural challenges, opportunities, and strategic directions for advancing the field.

6.2. Data analysis

6.2.1. Evolution and impact of ML on poverty mapping research

The advancement within a specific field of study can be observed in the evolution of academic papers, which is an important indicator of research trends. Upon conducting an in-depth research assessment, the area of machine learning for poverty mapping has emerged with notable trends and regional concentrations. The field of data-driven analysis is evolving through a combination of ML methods and conventional non-machine learning (non-ML) technologies. Long the foundation for

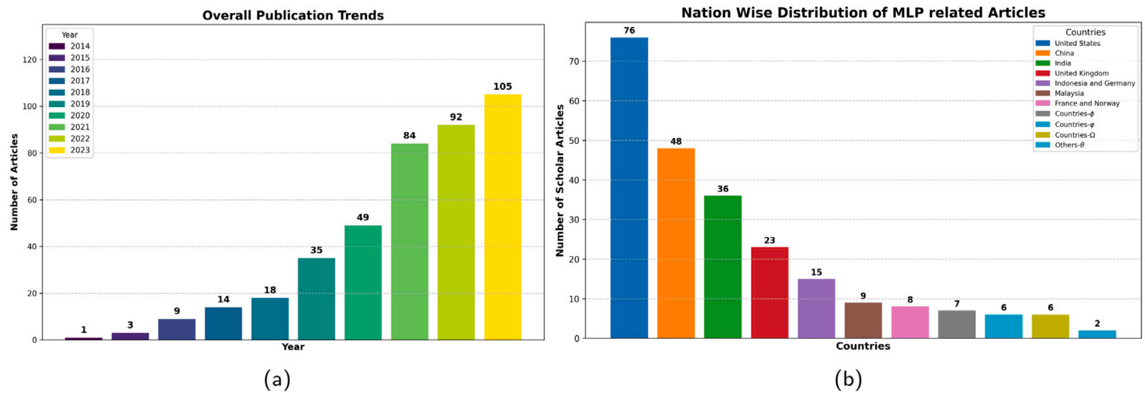

Fig. 7. (a): Overall Publication Trends in Years (b) Nation Wise Distribution of Publications of Related Articles.

poverty mapping initiatives, non-ML techniques like surveys, censuses, and remote sensing data offer insightful information but frequently run into issues with cost, time, and data quality. Conversely, ML approaches are transforming poverty mapping through the use of sophisticated algorithms to identify geographical correlations, efficiently validate models, and combine various datasets to produce more accurate predictions. Although machine learning might improve the accuracy of poverty mapping, it is important to recall that data quality is still the most important factor in producing trustworthy poverty maps.

A notable increase in the application of machine learning for poverty mapping is indicated by the Number of Articles Published. The Fig. 7(a) shows a striking rise in published publications from 1 in 2014 to 105 in 2023, indicating the field's increasing interest. The dispersion of publications among nations demonstrates a heterogeneous contribution landscape. This tendency can be attributed to several things, including the introduction of the Sustainable Development Goals (SDGs) program in 2015, which may have stimulated study into the reduction of poverty. Additionally, improvements in proper data-gathering techniques along with machine learning capabilities are important. Researchers are discovering that as algorithms get more advanced, they can be used more effectively to solve complicated problems including poverty mapping. Furthermore, machine learning's popularity may also be influenced by its ability to provide detailed insights into poverty on a micro level. Another factor contributing to machine learning's attraction to politicians and groups looking for more focused interventions is its capacity to offer in-depth insights into poverty at the micro level.

The evolution of academic papers within a specific field serves as a critical indicator of research trends and advancements. A comprehensive assessment of the literature reveals significant developments in the application of machine learning for poverty mapping, with notable regional and thematic trends. The integration of data-driven methodologies combines traditional non-ML techniques, such as surveys, censuses, and remote sensing data, with advanced ML approaches. While non-ML methods have long been foundational for poverty mapping, they often face challenges related to cost, time, and data quality. In contrast, ML techniques leverage sophisticated algorithms to uncover geographic correlations, validate models efficiently, and integrate diverse datasets, resulting in more accurate and actionable poverty maps. However, despite these advancements, data quality remains paramount in ensuring the reliability of poverty mapping outputs.

The growing prominence of ML in poverty mapping is evident in the sharp rise in publications within the field. The number of articles has increased from just 3 in 2015 to 105 in 2023, reflecting heightened research interest and activity. This surge can be attributed to several factors, including the introduction of the Sustainable Development Goals in 2015, which spurred global efforts to address poverty. Additionally, advancements in data collection techniques and the expanding capabilities of machine learning algorithms have further driven this growth. ML offers the ability to address complex challenges in poverty mapping by providing micro-level insights, enabling policymakers and organizations to design targeted interventions. The appeal of ML lies in its predictive accuracy with its potential to deliver nuanced, actionable insights, making it a vital tool for addressing poverty in diverse contexts.

The increasing volume of scholarly articles on poverty mapping using machine learning underscores a growing global commitment to leveraging advanced technologies for social good. This research effort, involving contributions from diverse nations, reflects a shared determination to combat poverty through innovative methodologies and data-driven strategies. The breadth of studies and international participation highlights a coordinated global initiative to harness machine learning for sustainable development and poverty alleviation.

Notably, countries such as the United States, China, and India have emerged as leaders in research output, driving the field's progression. As illustrated in Fig. 7(b), there is substantial worldwide engagement in using machine learning for poverty mapping, with notable contributions from various countries. However, a discernible disparity exists in the volume of research across regions, suggesting opportunities for enhanced collaboration and knowledge exchange to advance the discipline further. The bar graph (Fig. 7(b) provides a detailed breakdown of research contributions: United States leads with 76 articles, followed by China with 48, India with 36, the United Kingdom with 23, Indonesia and Germany with 15 each, Malaysia with 9, and France and Norway with 8 each. Additionally, Countries-ϕ (Thailand, the Netherlands, Australia, and Belgium) contributed 7 articles each; Countries-Ω (Italy, Spain, Bangladesh, Ghana, Mexico, and Sweden) contributed 6 each; and Countries-φ (Colombia, Saudi Arabia, France, and Chile) contributed 2 each. Lastly, others-θ (Romania, the Philippines, Jordan, Canada, Denmark, Singapore, and Brazil) contributed one article each. This global distribution, as depicted in Fig. 7, underscores the widespread interest in utilizing machine learning for poverty mapping while highlighting the need for broader participation and stronger international collaborations to foster innovation and equity in this critical area of research.

6.2.2. Research objectives and geographical distribution in poverty mapping

Understanding the objectives of ML applications in poverty mapping is essential for evaluating their impact on poverty research. By exploring these objectives, researchers can gain valuable insights into the potential implications of ML techniques in this domain. This approach facilitates a deeper understanding of how ML, in combination with traditional non-ML methods, contributes to advancements in poverty mapping and efforts to address socioeconomic inequalities. Table 3 summarizes 93 research objectives identified across 215 publications. Key objectives include assessing poverty levels (11 publications), mapping poverty using satellite imagery (47 publications), and examining

Table 3
Research Objectives and Applications of Machine Learning in Poverty Mapping.

The main aim of studies	Ref
Application of machine learning algorithms to PMT development improving the out-of-sample prediction	McBride and Nichols (2018), Kambuya (2020), Mcbride and Nichols (2015)
To predict poverty using machine learning	Zixi (2021), Monlezun et al. (2021), Sheng (2021), Shen et al. (2021), Verme (2020), Li et al. (2022), Kim (2021), Ramesh et al. (2021), Clark et al. (2021), Veres and Sampson (2021), Xiao (2021)
To examine relationship between migration factor and poverty	Zhang et al. (2021b), Martey and Armah (2021)
To cluster poverty data	Sano and Nindito (2016), Garg and Joshi (2021)
To identify poor households in data-scarce countries	Kshirsagar et al. (2017), Guaraca et al. (2021)
To monitor water level	Gallo et al. (2021), Manzione and Matulovic (2021)
To predict market prices	Zegarra and MacHicao (2021), Guo et al. (2021)
To examine temporal patterns in the relationship between jail cycling and community cases of COVID-1	Reinhart and Chen (2021)
To examine how different factors affect the intentional behavior towards community forestry among poor households	Mamun et al. (2020)
To Identify key determinants of a switch from biomass to non-biomass stoves	Neto-Bradley et al. (2020)
To identify the households that escaped and fell into poverty in the later time	Narendranath et al. (2018)
To investigate poverty relationship with socioeconomic predictors	Rachmawati and Pusponegoro (2018)

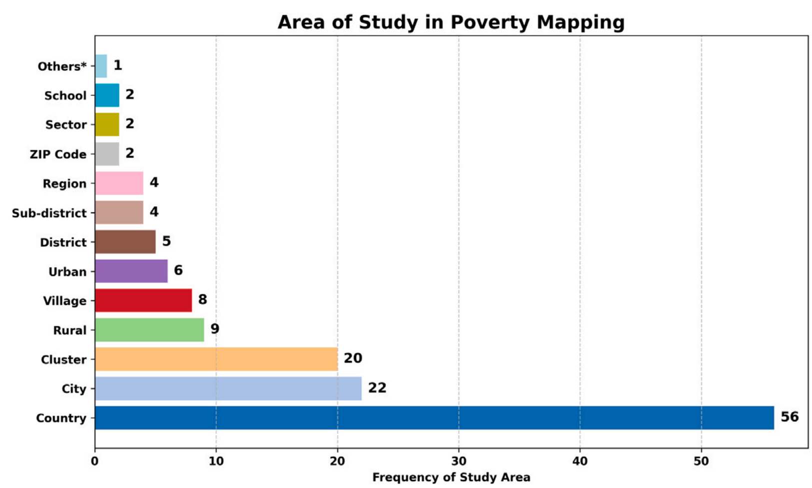

Fig. 8. Distribution of Focused Study Area for Poverty Mapping.

the relationship between socioeconomic factors and health issues (11 publications). A detailed summary of these studies is provided in Table 3, with additional information available in Appendix A.

Mapping poverty using satellite imagery has become increasingly common. Here the integration of ML with satellite data provides a more efficient and cost-effective alternative to traditional surveys and censuses, which are often labor-intensive. The availability of satellite images allows for rapid computational processing, enabling the extraction of features that represent poverty. Key indicators derived from these images include the nighttime light index, vegetation index, rainfall patterns, and soil surface temperature. These features serve as valuable proxies for socioeconomic conditions, enhancing our understanding of poverty distribution across different regions.

6.2.3. Topographical distribution for understanding poverty mapping

Understanding socio-economic disparities necessitates an examination of topographical distribution. To create comprehensive poverty maps, machine learning techniques are employed to analyze and interpret demographic data, satellite imagery, and geospatial information. By leveraging sophisticated algorithms, machine learning enhances the accuracy and depth of poverty mapping, providing valuable insights for targeted interventions and policy decisions. Researchers aim to quantify the relationship between socio-economic factors and health issues through quantitative evaluations. Accurately determining poverty levels enables governments to allocate subsidies more effectively, categorizing individuals, communities, regions, or entire nations based on their poverty status. Various datasets are utilized in these studies; for instance, the nighttime light index derived from satellite imagery serves as a reliable reference for estimating poverty levels. Additionally, significant socio-economic variables are sourced from the Demographic and Health Survey, while geographic data helps elucidate how economic challenges manifest in spatial contexts.

This study examines 131 locations across the globe, identified from 215 scholarly articles investigating the relationship between ML and poverty alleviation initiatives, as summarized in Table 4. Among these studies, China emerges as the most frequently analyzed country, serving as the focal point in 22 investigations. India follows with 17 studies, underscoring the growing significance of ML approaches in addressing poverty-related challenges. The United States ranks third, with 13 studies highlighting the contributions of its scientific community to advancing ML techniques for poverty reduction. The assessment of poverty severity across these studies utilized 21 distinct measures, as presented in Fig. 8. The majority of the research (56 studies) concentrated on national-level analyses, while 35 studies focused on poverty at the provincial level, comparable to county or state scales. Furthermore, 20 studies explored poverty at the cluster level, involving groups of households participating in surveys. A smaller subset of research

Table 4
Distribution based on Geo-location for Understanding Poverty Mapping.

Geolocation of study (with quantity)	Ref
China (22)	Li et al. (2019), Wu and Tan (2019), Deng et al. (2017), Peng et al. (2019), Lakner et al. (2022), Liu et al. (2021a), Wang (2021b), Zhang et al. (2021a), Zhao et al. (2021), Zhang et al. (2021b), Sheng (2021), Guo et al. (2021), Gao et al. (2021), Xu et al. (2021b), Wang et al. (2021b), Jiang et al. (2021), Li et al. (2021), Xu et al. (2021a), Yin et al. (2021), Liu et al. (2021b), Han et al. (2021)
India (17)	Tang et al. (2016), Pandey et al. (2018), Subash et al. (2018), Kim et al. (2021), Kherwa et al. (2021), Agarwal et al. (2019), Saini et al. (2020), Brahma and Mukherjee (2021), Urfels et al. (2021), Wang et al. (2021a), Narendranath et al. (2018), Singha et al. (2020), Neto-Bradley et al. (2020), Aizawa (2020), Rajagopal et al. (2021), Majumder et al. (2021), Debnath et al. (2021)
United States (13)	Irvin et al. (2020), Mukherjee et al. (2021), Dipnall et al. (2016), Chakrabarty and Biswas (2018), Tracy et al. (2021), Jendryke and McClure (2021), Wang (2021a), Dharmasena et al. (2016), Tiwari et al. (2021), Hase et al. (2020), E. and J. (2018), Platt et al. (2018), Farrokhvar et al. (2018)
Nigeria (11)	Wong et al. (2018), Jean et al. (2016), Gram-Hansen et al. (2019), Buddaraju et al. (2021), Okaidat et al. (2021), Crystal and Irvine (2019), Ni et al. (2020), Browne et al. (2021), Debnath et al. (2021), Kraamwinkel et al. (2019), Xiao (2021)
Africa (10)	Perez et al. (2017), Gram-Hansen et al. (2019), Yeh et al. (2020), Xie et al. (2016a), Shin et al. (2018), Li and Yao (2018), Khan and Blumenstock (2019), Botchey et al. (2020), Sheehan et al. (2019), Brandt et al. (2020)
Malawi (10)	El Naqa et al. (2015), Wong et al. (2018), Jean et al. (2016), Okaidat et al. (2021), Ni et al. (2020), Jarry et al. (2021), Oughton and Mathur (2021), Sohnesen and Stender (2017), Xiao (2021), Mcbride and Nichols (2015), Brandt et al. (2020), McBride and Nichols (2018)
Uganda (8)	Jean et al. (2016), Ayush et al. (2021), Hall et al. (2023), Ni et al. (2020), Ayush et al. (2020), Xie et al. (2016a), Browne et al. (2021), Sohnesen and Stender (2017)
Mexico (7)	Babenko et al. (2017), Salas et al. (2021), Rincón (2019), Calil et al. (2017), Rodriguez-barrios et al. (2021), nez et al. (2021)
Ghana (6)	Hall et al. (2023), Engstrom et al. (2019b), Browne et al. (2021), Martey and Armah (2021), Awoin et al. (2020), Kwarteng et al. (2021)
Ethiopia (6)	Kondmann and Zhu (2020), Oughton and Mathur (2021), Browne et al. (2021), Sohnesen and Stender (2017), Aizawa (2020), Xiao (2021)
Rwanda (6)	Jean et al. (2016), Kondmann and Zhu (2020), Buddaraju et al. (2021), Ni et al. (2020), Sohnesen and Stender (2017), Njuguna and McSharry (2017)
Bangladesh (6)	Mirelman et al. (2016), Zhao et al. (2019), Browne et al. (2021), Hossain et al. (2019), Steele et al. (2017), Shen et al. (2021)
Thailand (5)	Wolk et al. (2021), Sangkasem and Puttanapong (2020), Puttanapong et al. (2021), Simud et al. (2020), Kambuya (2020)
Mumbai (4)	Gram-Hansen et al. (2019), Wurm et al. (2019), Stark et al. (2019), Debnath et al. (2021)

analyzed localized scales, including districts (5 studies), sub-districts (4 studies), and villages (8 studies).

A detailed synthesis of these findings, along with the core research interests of the major studies, is provided in Appendix A, which includes a comprehensive chart summarizing the key contributions in this field.

6.2.4. Datasets and methodological approaches

Poverty mapping leverages a diverse array of datasets, ranging from traditional sources such as surveys and censuses to contemporary tools like satellite imagery and crowdsourcing platforms. The integration of these datasets with advanced ML methodologies facilitates the creation of highly precise and granular poverty maps. These maps play a pivotal role in enabling policymakers to make data-driven decisions and implement targeted interventions aimed at alleviating poverty. Prominent datasets utilized in poverty mapping include OpenStreetMap (OSM), geospatial data, Demographic and Health Surveys (DHS), and the Nighttime Light Index, as depicted in Fig. 9. For instance, the Nighttime Light Index is widely recognized as a proxy for estimating poverty, based on the premise that areas with greater artificial light intensity correlate with higher living standards. However, this approach has notable limitations in regions with minimal or no artificial lighting, where it fails to effectively capture poverty levels. DHS data provide crucial socioeconomic indicators, offering insights into poverty dynamics and their underlying factors. Geospatial data are also extensively employed to infer socioeconomic conditions through spatial analysis. Moreover, OSM serves as an invaluable resource, offering open-source, crowd-sourced geographic data that support the investigation of complex poverty-related issues. The distribution of these datasets based on the use ratio, as shown in Fig. 9, highlights their significance in advancing ML-based approaches to poverty mapping and analysis.

Researchers have utilized these datasets to map poverty using satellite imagery, explore the relationship between socioeconomic status and health, and quantify the severity of poverty. Among the dataset composition, 18% comprises geospatial data, which provides information on the Earth's surface and its spatial relationships. Data from the Demographic and Health Surveys, focusing on health-related variables and population demographics, account for 22% of the total. The largest share, 38%, is represented by the Nighttime Light Index, which measures artificial light emissions during nighttime as a proxy for economic activity and living standards. OpenStreetMap (OSM) contributes 8% with its open-source and modifiable geospatial data, while satellite imagery accounts for 15%, providing high resolution images from orbiting satellites. Detailed documentation of the datasets and their sources, as cited by various authors in their studies, is provided in Appendix B. Additionally, Appendix C presents a classification of the research papers based on various factors, including the datasets utilized and their major findings, offering a comprehensive overview of the contributions in this field.

6.2.5. Use of ML and AI in poverty mapping

AI systems employ ML algorithms to autonomously perform complex tasks, such as predicting outcomes based on input data. In poverty mapping, ML methodologies are categorized into three key types: supervised learning, unsupervised learning, and reinforcement learning, each tailored to address specific analytical requirements for poverty assessment and intervention. Supervised learning, widely utilized for classification and regression tasks, involves training models on labeled

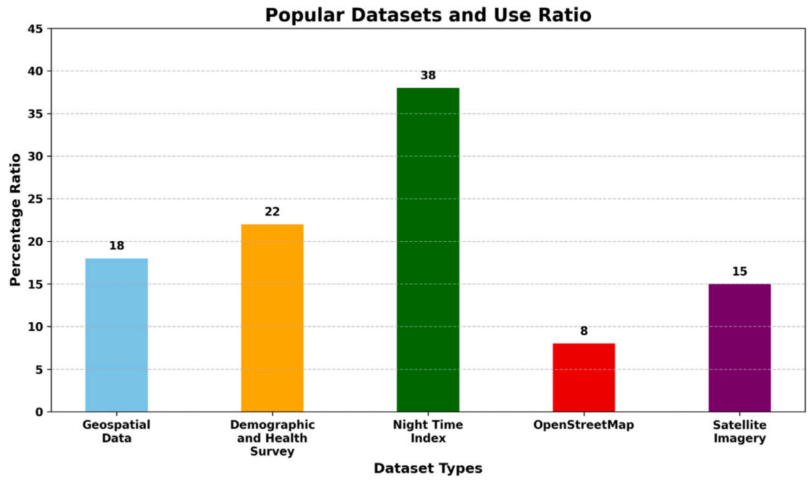

Fig. 9. Popular Datasets used in Poverty mapping.

datasets. This approach is instrumental in predicting poverty levels by leveraging labeled socioeconomic data, satellite imagery, and demographic indicators. By learning the relationships between input features and outputs, supervised learning enables accurate predictions for previously unseen data, facilitating the identification of poverty-stricken areas and supporting evidence-based decision-making.

Unsupervised learning, in contrast, focuses on analyzing unlabeled datasets to uncover hidden patterns or structures within the data. A notable application is clustering, which groups data points with similar attributes, providing valuable insights into communities with shared socioeconomic characteristics or poverty trends. Reinforcement learning introduces a transformative paradigm by enabling algorithms to learn optimal actions through iterative trial-and-error interactions with dynamic environments. This approach, as emphasized by its significant role in poverty mapping, is particularly effective in developing adaptive intervention strategies and resource allocation policies. By incorporating long-term impacts and diverse socioeconomic factors, reinforcement learning systems optimize resource distribution, adjust poverty alleviation strategies to evolving conditions, and determine the timing and sequencing of interventions for diverse communities, making it a pivotal tool for addressing the complexities of poverty assessment and intervention design. The use ratio of different ML models in research is presented in Fig. 10.

Since 2015, there has been a consistent increase in scholarly publications on ML applications in poverty research, with the majority of contributions originating from the United States. These studies primarily focus on mapping poverty using satellite imagery, examining the relationship between socioeconomic status and health, and characterizing poverty dynamics. Consequently, supervised learning techniques employ labeled datasets to develop predictive models that link inputs to outputs, while unsupervised learning methods explore hidden patterns in unlabeled data, expanding the scope of ML in addressing multifaceted challenges like poverty alleviation.

Regression, a statistical method used to examine the relationship between a dependent variable and one or more explanatory variables, is a fundamental technique for analyzing the complex interplay between poverty and socioeconomic factors, such as teen birth rates. By predicting the values of dependent variables based on explanatory variables, regression analysis provides critical insights that inform evidence-based policymaking and the design of targeted interventions. Similarly, classification, which involves organizing data into predefined categories (e.g., poverty levels), facilitates the identification of distinct patterns within poverty-related data. This approach enables precise interventions and supports well-informed policy decisions by highlighting unique characteristics and trends within the data. Clustering, another vital data analysis technique, involves grouping similar data

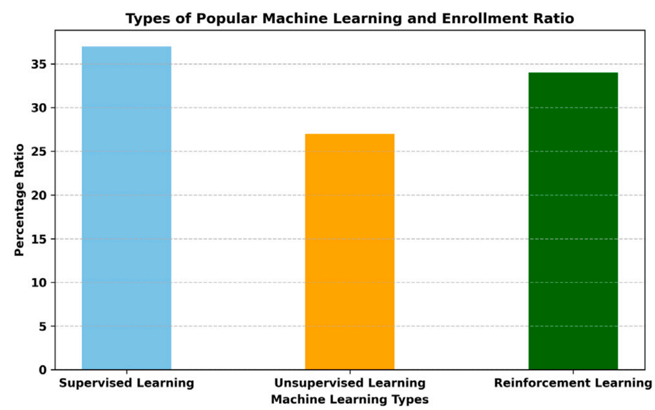

Fig. 10. Types of Popular Machine Learning Models and Enrollment in Poverty Mapping.

points into clusters. When applied to poverty-related datasets, clustering effectively identifies hotspots and socioeconomic trends, offering a deeper understanding of the root causes of poverty. This technique equips policymakers with the knowledge necessary to develop targeted measures that address specific issues within identified clusters, thereby enhancing the effectiveness of poverty alleviation strategies.

As depicted in the accompanying Fig. 11, the reviewed 215 research articles are categorized into seven thematic groups. Among these, 94 publications address regression-related challenges, while 87 focus on classification problems. Additionally, 12 papers explore clustering issues. Some studies intersect multiple categories: five papers examine both clustering and regression, another five address clustering and classification, three investigate clustering, classification, and regression simultaneously, and nine combine regression and classification. A detailed analysis of these categories, along with their findings and implications, is presented in Appendix C.

6.2.6. *Algorithmic approaches of ML in poverty mapping*

ML algorithms are crucial for poverty mapping, enabling the analysis of complex datasets to discern patterns that guide targeted interventions. They excel in classification, regression, and clustering tasks, each contributing uniquely to the understanding of poverty dynamics. Classification methods effectively categorize regions by poverty levels using data from satellite imagery and household surveys. Regression models estimate socioeconomic indicators, such as income and access to services, based on predictive variables. Clustering techniques group

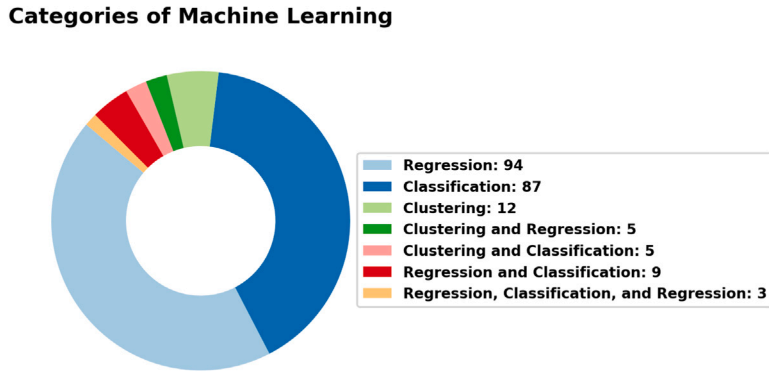

Fig. 11. Enrollment Categories of Machine Learning used in Poverty Mapping.

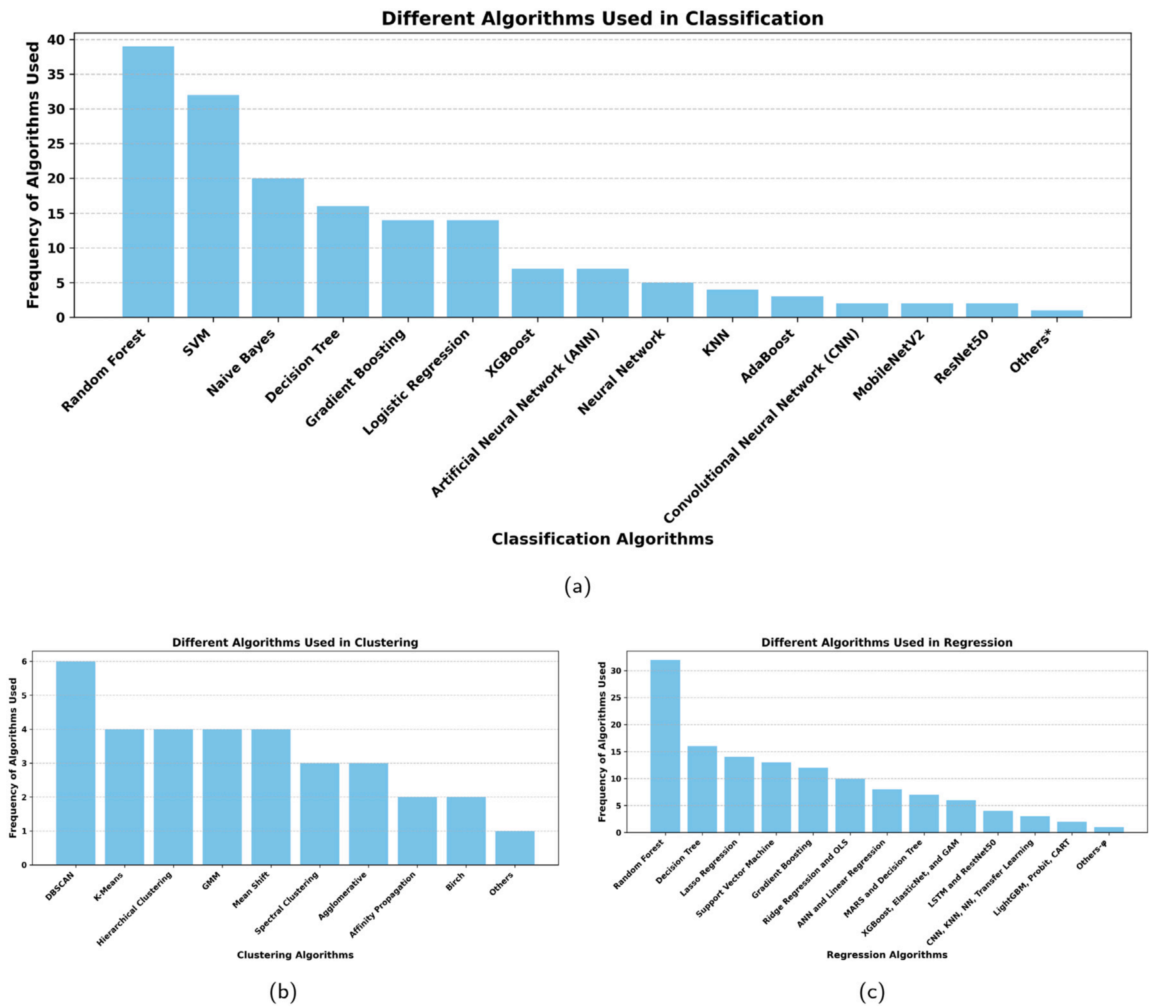

Fig. 12. (a): Algorithms Used in Classification (b) Algorithms Used in Regression (c) Algorithms Used in clustering in Poverty Mapping.

regions with similar socioeconomic traits, aiding resource allocation and policy formulation. The choice of algorithms depends on data characteristics, analysis scale, and study objectives, as no single method is universally optimal. Supervised learning techniques, including decision trees and support vector machines, are prevalent for predictive tasks, while unsupervised methods like k-means clustering identify structures in unlabeled data. We have compiled a list of commonly utilized algorithms in machine learning for poverty applications. The Figs. 12 outlines specific model selections for classification, regression, and clustering tasks in this domain.

The following provides analysis of machine learning classification methods: Random Forest is the most frequently utilized algorithm, appearing in 39 journal articles, followed by SVM with 32 occurrences. Naive Bayes is cited in 20 articles, while Decision Tree and Gradient Boosting are referenced 16 and 14 times, respectively. Both ANN and Logistic Regression are noted in 14 articles each. Neural Network and XGBoost appear five and seven times, respectively. CNN, MobileNetV2, and ResNet50 are each employed twice, while Adaboost is used three times. The dataset also includes four instances of KNN and one instance for each of the algorithms categorized as other-Ω

(including Self-Organizing Feature Maps (SOFM), K-Medoids Clustering, CLIQUE, Bisecting K-Means, Factorization Machines Clustering, Affinity Clustering, Isotonic Regression, and Poisson Regression), as depicted in Fig. 12(a). Random Forest remains a popular choice in poverty mapping due to its simplicity and interpretability; however, it is not universally superior across all contexts. Advanced methods such as gradient boosting demonstrate exceptional performance in capturing complex, non-linear relationships, while deep learning techniques offer unparalleled scalability and accuracy in data-rich scenarios. Each modeling approach presents unique strengths and limitations, emphasizing the need for a context-specific selection aligned with the objectives, data characteristics, and computational resources of the application.

In the context of clustering methods, K-Means appears in 4 journal articles, while DBSCAN is used in 6 instances. Hierarchical Clustering and Gaussian Mixture Model (GMM) are each referenced 4 times. Mean Shift is recorded 4 times, and both Agglomerative Clustering and Affinity Propagation occur twice each. Birch is mentioned twice, and spectral clustering appears three times overall. These counts provide insight into the distribution of various clustering approaches within the dataset. Additionally, other methods SOFM, Chinese Restaurant Process (CRP) Clustering, CLIQUE, Bisecting K-Means, Factorization Machines Clustering, Affinity Clustering, and Isotonic Clustering) are noted once each. The overall distribution is illustrated in Fig. 12(b).

The distribution of regression approaches utilized is as follows: Random Forest appears in 32 journal articles, followed by Decision Tree with 16 instances. Lasso Regression is cited 14 times, while SVM appears in 13 articles and Gradient Boosting in 12. Both Ridge Regression and Ordinary Least Squares (OLS) are present in 10 instances each. ANN and Linear Regression are utilized 8 times, while Multivariate Adaptive Regression Splines (MARS), Multiple Linear Regression, and Decision Trees each occur 7 times. XGBoost, Elastic Net, and Generalized Additive Model (GAM) are each implemented in 6 cases, while LSTM and ResNet50 are used 4 times. KNN, CNN, Neural Network, Transfer Learning, Bayesian Regression, and Generalized Linear Models (GLM) appear 3 times each. Other methods categorized as others-ϕ (including Huber Regression, Principal Component Regression (PCR), Kernel Ridge Regression, Isotonic Regression, Quantile Regression, and Poisson Regression) are noted once each. This overall distribution is illustrated in Fig. 12(c), with expanded information available in Appendix B.

6.2.7. *Evaluation metrics grouped by the purpose of studies*

Evaluation metrics in research projects utilizing machine learning algorithms to enhance Poverty Measurement Tools predominantly focus on the predictive accuracy of models. Key metrics for assessing model performance include Mean Squared Error, Root Mean Squared Error, Mean Absolute Error, R-squared, and cross-validation metrics. A diverse array of datasets is employed, ranging from mobile phone data to social media and satellite imagery, with evaluation metrics encompassing standard classification measures such as accuracy, precision, recall, and F1-score, as well as spatial metrics pertinent to mapping tasks.

Given the complexity of these analyses, statistical measures like correlation coefficients and significance tests are emphasized in studies examining the relationships between migration factors, education, and poverty. The selection of appropriate evaluation metrics is contingent upon the specific objectives and characteristics inherent in each study, ensuring a comprehensive assessment aligned with the distinct aims of the research. Appendix C provides a detailed overview of the evaluation metric scores and model analysis tools, along with comparative results across various categories and a comprehensive source and reference list.

7. Challenges and future opportunities

ML techniques have emerged as transformative tools in poverty mapping, leveraging satellite data and advanced algorithms to provide granular insights into micro-geographical poverty patterns. By identifying these patterns with a precision that traditional methods cannot achieve, ML models, such as Random Forest, have proven invaluable in enhancing the accuracy and interpretability of poverty assessments. These advancements offer policymakers critical insights for targeted interventions and more effective resource allocation. A review of diverse studies underscores the potential of ML in reshaping poverty mapping and driving strategies that support equitable development. However, applying ML must account for contextual socio-economic factors, as reliance solely on satellite imagery may overlook nuances inherent in local environments.

While nighttime lights can provide valuable insights into economic activity, their utility in poverty mapping is limited when used in isolation. They are less effective in regions with sparse artificial lighting, such as rural or impoverished areas, where significant economic or social activities may remain undetected. To address these gaps, nighttime lights should be used in conjunction with other data sources—such as high-resolution satellite imagery, socio-economic surveys, and infrastructure maps—for a more comprehensive understanding of poverty dynamics. While nighttime light indices are valuable for poverty estimation due to their correlation with economic activity, their predictive power is significantly enhanced when combined with daytime features such as land cover classification, building information, and road networks, which address limitations like rural underrepresentation and urban signal saturation. Integrating these datasets allows analysts to capture hidden or less visible aspects of poverty, such as informal economies, resource accessibility, and population distribution, thereby improving the accuracy and relevance of poverty mapping efforts. The integration of NTLI with daytime features provides complementary insights into socioeconomic factors, enabling more accurate and granular poverty assessments by capturing dimensions like agricultural practices, infrastructure quality, and economic connectivity.

Despite its promise, ML-based poverty mapping faces significant challenges that must be addressed to maximize its potential. Key issues include biases in training datasets, which can lead to inaccurate representation of poverty levels, and the inherent limitations of remote sensing technologies in capturing socio-economic intricacies. Furthermore, the quality and availability of large-scale, reliable datasets remain a bottleneck, especially in resource-constrained regions. These limitations highlight the importance of integrating qualitative data with ML methodologies to develop a more comprehensive understanding of poverty dynamics. Future research should also explore emerging technologies such as artificial intelligence and big data analytics, which have the potential to refine poverty mapping and enhance its scalability.

Collaborations among researchers, policymakers, and local communities are essential to fully leverage ML for poverty mapping. Such partnerships can address critical concerns like data equity, processing capacity, and methodological relevance while fostering innovation. By merging interdisciplinary expertise, stakeholders can create robust, efficient, and contextually appropriate frameworks. The strategic application of ML to poverty mapping aligns with SDGs and holds the potential to reduce global poverty through informed decision-making and targeted interventions significantly. Regional variations in research output highlight disparities in data access and methodological focus, suggesting the need for more inclusive and globally representative datasets to enhance the generalizability of poverty-mapping models.

Utilizing daytime imagery enables researchers to capture more relevant features, enhancing the training and performance of machine learning models in poverty mapping. Addressing data availability constraints is equally crucial, as the quality and diversity of datasets directly influence the robustness of these applications. Access to high-resolution datasets fosters more comprehensive analyses, driving the development of accurate poverty mapping strategies and empowering policymakers to implement more effective poverty alleviation efforts.

Recognizing the "no free lunch" theorem is vital, as no single ML algorithm universally outperforms others across all scenarios. This underscores the importance of selecting and customizing algorithms for specific poverty-mapping contexts. By systematically addressing these challenges and fostering creativity in ML applications, we can transform poverty mapping into a powerful instrument for global development. Ongoing research, combined with sustained investment in data and technology, will ensure that ML continues to evolve as a critical tool in the fight against poverty.

8. Conclusion

This paper presents a comprehensive review of the expanding applications of ML in poverty mapping, emphasizing critical elements such as research objectives, emerging trends, datasets, algorithms, and evaluation metrics. Synthesizing findings from diverse studies highlights the significant role of ML in enhancing poverty assessments and facilitating targeted interventions. Furthermore, the review underscores the critical role of community engagement and ML integration in achieving sustainable poverty alleviation. While ML demonstrates significant potential in refining poverty mapping methodologies, addressing inherent challenges is crucial to ensure fair and impactful outcomes. Notably, the growing adoption of ML techniques in countries such as the United States, China, and India underscores the global relevance of this research. These studies have primarily focused on leveraging satellite data to map poverty, exploring the relationship between socio-economic conditions and health outcomes, and assessing the prevalence of poverty with increased precision.

A key strength of these studies lies in their reliance on publicly accessible datasets, such as the Nighttime Light Index and Demographic Surveys, which foster innovative experimentation. Future studies should prioritize the integration of daytime imagery alongside nighttime light data to enhance predictive accuracy. Additionally, there is a need for improved accessibility to geolocated training and evaluation datasets, which will facilitate more robust machine learning applications. Supervised and unsupervised learning techniques dominate the field, with widely employed algorithms like naïve Bayes, random forests, and support vector machines. Random Forest, while widely adopted for its interpretability and robust performance, is often outperformed by methods like gradient boosting and deep learning, which deliver higher predictive accuracy for complex datasets at the cost of increased computational demands. Our analysis underscores the limitations of nighttime light indices when used in isolation, highlighting the value of combining them with daytime features like land cover and building information to achieve more accurate and granular poverty assessments. Task-specific evaluation metrics, such as mean squared error for regression tasks and F1 scores for classification problems, underline the rigorous validation of these methodologies.

The proliferation of research from 2014 to 2023 highlights the transformative potential of technology in addressing poverty and empowering marginalized communities. However, this progress is accompanied by pressing challenges, particularly ethical concerns such as algorithmic biases and data privacy. Addressing these issues is essential for fostering sustainable and equitable poverty alleviation initiatives. Incorporating community feedback and blending ML approaches with conventional methods are imperative to ensure contextually sensitive and inclusive solutions.

The future of ML in poverty mapping is promising but contingent upon overcoming these critical challenges. Ethical safeguards, interdisciplinary collaboration, and continued innovation will be essential to unlocking its full potential. We recommend further research into hybrid models incorporating traditional survey data and innovative remote sensing techniques. This could enhance the robustness of poverty assessments, particularly in underrepresented regions. Similarly, we recommend adopting a framework that assigns weights to included studies based on their methodological rigor, findings' robustness, and societal impact. This would provide a more nuanced understanding of the reliability and significance of individual studies, addressing the limitations of equal weighting. Addressing these limitations, ML can evolve into a transformative tool for achieving global development goals and advancing socio-economic equity.

CRediT authorship contribution statement

Badri Raj Lamichhane: Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **Mahmud Isnan:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Teerayut Horanont:** Writing – original draft, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors have no conflicts of interest to declare. All co-authors have seen and agree with the contents of the manuscript and there is no financial and non-financial interest to report. We certify that the submission is original work and is not under review at any other publication.

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Appendix A

See Table A.1.

Appendix B

See Table B.1.

Table A.1
The main aims of studies (contd...).

The main aim of studies	Ref
To examine the relationship between poverty and the impact of schistosomiasis	Rinaldo et al. (2021)
To examine the relationship between various neighborhood characteristics and the change in average household income in the neighborhood	Hipp et al. (2017)
To analyze the key characteristics of low-income groups	Deng et al. (2017)
To analyze the relationship between the capability of the poor population and poverty	Sang and Guo (2021)
To assess coastal risks	Calil et al. (2017)

(continued on next page)

Table A.1 (*continued*).

The main aim of studies	Ref
To assess the performance of rural banks	Awoin et al. (2020)
To automatically detect electric transmission infrastructure from aerial imagery	Hu et al. (2020)
To carry out analysis of both yield and gross margin gaps	Devkota and Yigezu (2020)
To classify user behavior using e-commerce data	Chen and Lian (2021)
To detect cavities	Sonavane et al. (2021)
To determine which rich countries are helping poorer ones the most	Sianes et al. (2014)
To diagnose covid-19	Delafiori et al. (2021)
To do a sentiment analysis of the two types of vaccines based on the Twitter platform	Nurdeni et al. (2021)
To Enhance Governance Issues through Opinion Extraction	Shaukat et al. (2020)
To estimate energy consumption	González-Vidal et al. (2017)
To explore the suite of risk factors that influence the transmission of Lymphatic Filariasis	Kwarteng et al. (2021)
To find risk factors for suicide	Berkelmans et al. (2021)
To forecast the catch volumes of fisheries	nez et al. (2021)
To determine the welfare candidate eligibility	Zakaria et al. (2016)
To predict market prices	Guo et al. (2021)
To predict economic development using Geolocation area	Sheehan et al. (2019)
To quantify the interdependence of the poverty and inequality goals	Lakner et al. (2022)
To examine the relationship between income distribution and economic development	Dutt and Tsetlin (2021)
To examine relationship between donation and socioeconomic	Farrokhvar et al. (2018)
To examine the relationship between land use and declining primate populations	Zhao et al. (2021)
To examine relationships between population densities and geographical factors	Nieves et al. (2017)
To analyze reading comprehension at elementary school	Rodriguez-barrios et al. (2021)
To examine the association Between commute time and poverty	Platt et al. (2018)
To examine the relationship between poverty and energy	Debnath et al. (2021)
To improve the performance of precision poverty alleviation	Wang (2021b)
To model the livelihood vulnerability state (LVS)	G.D.Singh (2021)
To monitor social-distancing and face mask	Sanjay et al. (2021)
To monitor trees outside of forests	Brandt et al. (2020)
To overcome over-indebtedness and fight poverty	Ferreira et al. (2021)
To predict flood using machine learning	Gauhar et al. (2021)
To predict poverty using government data	Loukis et al. (2020)
To predict renewable energy	Naik et al. (2021)
To quantify treatment effects of two improved living standards	DeFries et al. (2021)
To estimate the heterogeneous treatment effects of maternal agency	Kraamwinkel et al. (2019)
To identify spatial poverty determinants in rural	Liu et al. (2021a)

Table B.1
Datasets and their sources used by authors for learning in poverty mapping.

No	Datasets	Sources	Ref
1.	Nightlight (NTL) data	(DMPS) https://www.ospo.noaa.gov/Operations/DMSP/index.html, (VIIRS) https://www.earthdata.nasa.gov/learn/find-data/near-real-time/viirs https://earthengine.google.com/	– Jean et al. (2016), Subash et al. (2018), Li et al. (2019), Kondmann and Zhu (2020), Ivan et al. (2020), Xu et al. (2021b), Wang et al. (2021b), Jiang et al. (2021), Lin Htet et al. (2021), Wolk et al. (2021), Xu et al. (2021a), Kherwa et al. (2021), Ni et al. (2020), Sangkasem and Puttanapong (2020), Jarry et al. (2021), Saini et al. (2020), Yin et al. (2021), Xie et al. (2016a), Han et al. (2021), Tingzon et al. (2019), Puttanapong et al. (2021), Zhao et al. (2019), Njuguna and McSharry (2017), Sheehan et al. (2019)
2.	Demographic and Health Survey (DHS)	https://dhsprogram.com/data/	–
3.	OpenStreetMap (OSM)	https://www.openstreetmap.org/	Engstrom et al. (2019a), Wang et al. (2021b), Lin Htet et al. (2021), Yin et al. (2021), Piaggesi et al. (2019), Tingzon et al. (2019), Puttanapong et al. (2021), Zhao et al. (2019), Yuan et al. (2018), Kwarteng et al. (2021)
4.	Daytime Satellite Imagery	(Google Static Map) https://developers.google.com/maps/documentation/maps-static/overview	Jean et al. (2016), Pandey et al. (2018), Wolk et al. (2021), Kherwa et al. (2021), Buddaraju et al. (2021), Agarwal et al. (2019), Ni et al. (2020), Xie et al. (2016a), Tingzon et al. (2019),
5.	Digital globe	https://www.digitalglobe.com/company/about-us/	Babenko et al. (2017), Gram-Hansen et al. (2019), Ayush et al. (2021), Chao et al. (2021), Piaggesi et al. (2019), Ayush et al. (2020), Tingzon et al. (2019), Brandt et al. (2020)

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No	Datasets	Sources	Ref
6.	Living standard measurement surveys (LSMS)	https://www.worldbank.org/en/programs/lsms	Jean et al. (2016), Xiao (2021), Khaefi et al. (2019), Ayush et al. (2020), Jarry et al. (2021), Ayush et al. (2021), Yeh et al. (2020)
7.	Sentinel-1 and 2	https://sentinel.esa.int/web/sentinel/missions/sentinel-1, https://sentinel.esa.int/web/sentinel/missions/sentinel-2, https://earthengine.google.com/	Gram-Hansen et al. (2019), Jiang et al. (2021), Crystal and Irvine (2019), Hersh et al. (2020), Tingzon et al. (2020)
8.	Call detail Records (CDRS)	N/A	Hernandez et al. (2017), Khaefi et al. (2019), Khan and Blumenstock (2019), Aiken et al. (2020), Engstrom et al. (2019a), Njuguna and McSharry (2017), Steele et al. (2017)
9.	United states census bureau	https://www.census.gov/	Shin et al. (2018), Botchey et al. (2020), E. and J. (2018), Farrokhvar et al. (2018), Hipp et al. (2017)
10.	Landsat 8	https://landsat.gsfc.nasa.gov/satellites/landsat-8/, https://earthexplorer.usgs.gov/	Wu and Tan (2019), Prasetyo et al. (2021), Jiang et al. (2021), Mayani and Itagi (2021), Zhang et al. (2021b)
11.	Landsat 7	https://www.usgs.gov/landsat-missions/landsat-7, https://earthengine.google.com/	Perez et al. (2017), Salas et al. (2021), Zhang et al. (2021a,b)
12.	Land use data	http://www.unc.edu/~nielsen/data/data.htm, https://www.intelligence-airbusds.com/imagery/constellation/pleiades/	Wang et al. (2021b), Georganos et al. (2019), Zhao et al. (2019), Hipp et al. (2017)
13.	Economic Survey China data	https://www.chinayearbooks.com/2017/01	Wu and Tan (2019), Xu et al. (2021b), Wang et al. (2021b), Gao et al. (2021)
14.	Land surface temperature	https://neo.sci.gsfc.nasa.gov/view.php?datasetId=MOD11C1_M_LSTDA, https://earthengine.google.com/	Wong et al. (2018), Bakar et al. (2020), Browne et al. (2021), Zhang et al. (2021a)
15.	Quickbird-2	https://earth.esa.int/eogateway/missions/quickbird-2	Engstrom et al. (2017b), Wurm et al. (2019), Stark et al. (2019)
16.	Flood and drought data	https://iciwarm.info/abc/	Kraamwinkel et al. (2019)
17.	Planet	https://www.planet.com/	Babenko et al. (2017), Xiao (2021), DeFries et al. (2021)
18.	Worldview-2	https://earth.esa.int/eogateway/missions/worldview-2	Engstrom et al. (2019a), Brandt et al. (2020), Yuan et al. (2018)
19.	National Statistical Office of Thailand	http://www.nso.go.th/sites/2014en	Sangkasem and Puttanapong (2020), Puttanapong et al. (2021), Kambuya (2020)
20.	CHIRPS rainfall	https://www.chc.ucsb.edu/data/chirps, https://earthengine.google.com/	Hersh et al. (2020), Puttanapong et al. (2021), Rajagopal et al. (2021)
21.	Modis imagery	https://modis.gsfc.nasa.gov/	Hersh et al. (2020), Wang et al. (2021a)
22.	MCS-ENIGH household survey	https://en.www.inegi.org.mx/programas/mcs/2014/	Babenko et al. (2017), Rincón (2019)
23.	Normalized difference vegetation index (NDVI)	https://neo.sci.gsfc.nasa.gov/view.php?datasetId=MOD_NDVI_M, https://earthengine.google.com/	Wong et al. (2018), Puttanapong et al. (2021)
24.	E-kasih	https://www.kpwkm.gov.my/kpwkm/index.php?r=portal/aboutid=S2FXUVBnbTY1T0twYXhRcy8wcVl6Zz09	Sani et al. (2018), Bakar et al. (2020)
25.	Shuttle Radar Topography Mission (SRTM) digital elevation model (DEM) data	https://www.usgs.gov/centers/eros/science/usgs-eros-archive-digital-elevation-shuttle-radar-topography-mission-srtm-1	Wang et al. (2021b), Xu et al. (2021a)
26.	Landsat 5	https://landsat.gsfc.nasa.gov/satellites/landsat-5/	Couchoro et al. (2021), Zhang et al. (2021b)
27.	Google Earth (GE) imagery	https://earth.google.com/	Liu et al. (2021a)
28.	Afrobarometer	http://http//afrobarometer.org	Ni et al. (2020), Sheehan et al. (2019)
29.	Geoeye 1	https://earth.esa.int/eogateway/missions/geoeye-1	Engstrom et al. (2019a), Brandt et al. (2020)
30.	Thai People Map and Analytics Platform (TPMAP)	https://www.tpmap.in.th/	Puttanapong et al. (2021), Simud et al. (2020)
31.	Soil dataset	http://data.cma.cn/	Zhang et al. (2021a), Rajagopal et al. (2021)
32.	Encuesta de Hogares (EH)	https://www.ine.gob.bo/index.php/estadisticas-sociales/vivienda-y-servicios-basicos/encuestas-de-hogares-vivienda/	McBride and Nichols (2018), Calil et al. (2017)
33.	Second integrated household survey (ihs2)	https://microdata.worldbank.org/index.php/catalog/2307	McBride and Nichols (2018), Calil et al. (2017)
34.	Timor Leste living survey (tlls)	https://www.statistics.gov.tl/category/survey-indicators/timor-leste-survey-living-standards/	McBride and Nichols (2018), Calil et al. (2017)
35.	Poverty data taken from Badan Pusat Statitik (BPS)	https://www.bps.go.id/	Pangestu et al. (2020), Sano and Nindito (2016)

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No	Datasets	Sources	Ref
36.	United states geological survey	https://www.usgs.gov/	Puttanapong et al. (2021)
37.	Urban areas boundaries	http://www.rulamahue.cl/mapoteca/presentaciones/	Piaggesi et al. (2019)
38.	Multi-dimensional poverty Index (MPI)	http://www.statistics.gov.rw/publication/rphc4-main-indicators-report	Njuguna and McSharry (2017)
41.	USDA food environment atlas	https://www.ers.usda.gov/data-products/food-environment-atlas/	Dharmasena et al. (2016)
42.	Center for Disease Control and Prevention	https://www.cdc.gov/	Dharmasena et al. (2016)
43.	United States Bureau of Labor Statistics	https://www.bls.gov/	Dharmasena et al. (2016)
44.	USDA Food	https://www.usda.gov/topics/food-and-nutrition	Dharmasena et al. (2016)
45.	The National Health and Nutrition Examination Survey (NHANES)	http://wwwn.cdc.gov/Nchs/Nhanes/Search/nhanes09_10.aspx	Dipnall et al. (2016)
46.	Health and demographic surveillance system	https://www.icddrb.org/	Mirelman et al. (2016)
47.	Cumulated cyclone winds	https: //preview.grid.unep.ch/index.php?preview=aboutcat=2lang=eng	Wilson et al. (2017a)
48.	Gini coefficient of inequality	https://doi.org/10.1111/ssqu.12295	Wilson et al. (2017a)
49.	Infant mortality rates (IMR)	http://dx.doi.org/10.7927/H4PZ56R2	Wilson et al. (2017a)
50.	Housing and neighborhood quality survey data	https://innovatememphis.com/propertyhub/	Shin et al. (2018)
51.	County-level socioeconomic statistical data 2016	http://www.stats.gov.cn/tjsj/ndsj/2017/indexeh.htm	Xu et al. (2021a)
52.	Severe energy poverty index	https://www.ceew.in/access-survey	Wang et al. (2021a)
53.	Finer Resolution Observation and Monitoring Global land cover	https://www.dess.tsinghua.edu.cn/info/1120/1418.html	Xu et al. (2021a)
54.	Urban areas boundaries	http://www.rulamahue.cl/mapoteca/presentaciones/	Piaggesi et al. (2019)
55.	Housing and neighborhood quality survey data	https://innovatememphis.com/propertyhub/	Shin et al. (2018)

Table C.1
Comparisons of evaluation metrics grouped by the purpose of studies.

Ref	Datasets	Model/ analysis	Metric	Highest value
A.Application of ML to PMT development improving sample prediction				
Mcbride and Nichols (2015)	Encuesta de Hogares (EH), Second Integrated Household Survey (IHS2), Timor Leste Living Survey (TLLS)	RF	Total accuracy, Poverty accuracy, Balanced poverty accuracy criterion	Accuracy: 71.75–83.65
McBride and Nichols (2018)	Encuesta de Hogares (EH), Second Integrated Household Survey (IHS2), Timor Leste Living Survey (TLLS)	Least squares, Quantile regression, Poverty accuracy, Under coverage, Leakage, and Balanced poverty accuracy	Accuracy	74%–83%
Kambuya (2020)	Socioeconomic survey (SES)	Stepwise, Lasso, RF	Total accuracy, Poverty accuracy, Inclusion error, Exclusion error	Total accuracy:84.59–89.11
B. To estimate poverty using cell phone data				
Hernandez et al. (2017)	1. Guatemala's National Living Condition Survey (ENCOVI), 2. 200 population and Housing Cencuses, 3. Call detail record (CDR)	SVM, RF, Stochastic gradient boosting, Gaussian Mixture, Supervised Topic Models	R2, F1	Municipio-level (R2:0.76 and F1:0.84), urban(R2:0.69 and F1:0.73), rural(R2:0.46 and F1:0.59)
Khaefi et al. (2019)	1. DHS and LSMS data, 2. Call Detail Records 2018	SVM, Naïve Bayes, Elastic Net, Neural Network, Decision tree	R2, F1	R2:0.13, F1:0.68 and 0.7
Khan and Blumenstock (2019)	Mobile phone Call Detail Records (CDR)	Node2vec, Deepwalk, and LINE, GCN	Accuracy	43.2–81.1
Aiken et al. (2020)	1. Ultra-Poor (TUP) program, 2. Per capita monthly consumption 3. Asset-based wealth index 4. Food security index, 5. Financial inclusion index 6. A psychological well-being index for the primary woman	Logistic regression, RF, gradient boosting model	AUC	0.68–0.78

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Table C.1 (*continued*).

Ref	Datasets	Model/ analysis	Metric	Highest value
Oughton and Mathur (2021)	1. Malawian fourth integrated household survey 2. Ethiopian socioeconomic survey	VGG	R2	0.25–0.40
C. To estimate poverty using random forest				
Zhao et al. (2019)	1.DHS data 2.National Polar-orbiting Partnership 3. Nighttime light (NTL) 4.OSM data	RF, Linear regression, Transfer learning model	R2	0.58–0.70
Bakar et al. (2020)	Ekasih	Decision tree,	RF Accuracy	98%–99%
Tracy et al. (2021)	1.A retrospective review of patients undergoing abdominal wall herniorrhaphy 2013–2015, 2.A Distressed Communities Index(DCI)	RF	AUC	0.81–0.82
D. To map poverty using mobile phone and satellite data				
Njuguna and McSharry (2017)	1. Multi-dimensional poverty Index (MPI) 2. Call detail Records (CDRS), Nightlight satellite data, 3. Landscan population data	Lasso regression	Correlation, p-value	Correlation: 0.88
Steele et al. (2017)	DHS data, FII survey and national household surveys, CDR data,	RS data Bayesian geostatistical models (BGMS)	R2 and RMS	R2: 0.00–0.78
E. To map poverty using satellite data				
Jean et al. (2016)	1. Daytime satellite imagery, 2. LSMS data 3. DHS data 4. Nightlight data	VGGF	R2	0.55–0.75
Xie et al. (2016b)	1. DHS data, 2. NTL data, 3. LSMS data	VGGF	Accuracy, F1,	Accuracy: 0.52–0.78
Engstrom et al. (2017a)	1. Household Income and Expenditure Survey (HIES), 2. High Spatial Resolution Imagery (HSRI)	Lasso regression	R2	10% poverty line: 0.61, 40% poverty line: 0.61, average consumption: 0.60
Subash et al. (2018)	1. Night light data, 2. Data on rural poverty estimates at state level (goi)	ANN	R2 and RMSE	R2:0.56 and 0.59
Salas et al. (2021)	The residential blocks Landsat-7	Lenet-5, Resnet, ResNext, EfficientNet	ROC, precision–recall	ROC:0.81–0.84
Xiao (2021)	1. LSMS Survey, 2. Satellites data	RF, VGG11	R2	0.34–0.58
F. To map poverty using satellite data, social media, and geospatial information				
Tingzon et al. (2019)	1. DHS data, 2. Nighttime Lights, 3. Daytime Satellite Imagery, 4. Human Settlement Data (HRSL)	Transfer learning, RF	R2	0.49–0.63
Ledesma et al. (2020)	1. DHS data, 2. Social media advertising data, 3. Nighttime luminosity, 4. Daytime and nighttime land surface temperature, 5. NDVI, 6. Point of interest	Lasso regression, Ridge regression, RF, and lightGBM.	R2, MAE, C-statistic	0.00–0.66
G. To examine the relationship between poverty and education				
Hurst et al. (2020a)	Anonymized residential properties over an 18-month	Decision tree, Decision forest, Decision jungle, Neural network, SVM and Bayes point machine	Accuracy, precision, recall, F1,	Accuracy, 0.73–0.74
Hurst et al. (2020b)	The smart gas meter data	Decision trees	Accuracy, precision, recall,	Accuracy:0.59–0.67
H. To monitor poverty using gas smart meter				
Rengifo et al. (2018)	Censuses of the Republic of Chile	Decision trees	Accuracy and kappa	accuracy:0.87 for 1920 and 0.76 for 1930
Han et al. (2021)	1. Statistical Yearbook, and Education Statistical Data, 2. GDP indicator, 3. NTL Data	RF	Accuracy, kappa	accuracy:97.10–96.66%

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Table C.1 (*continued*).

Ref	Datasets	Model/ analysis	Metric	Highest value
I. To examine relationship between poverty and money transactions				
Engelmann et al. (2019)	Mass CDR and M-Money datasets	RF	F1, precision,	F1:0.63–0.71
Botchey et al. (2020)	Transactional logs from a mobile money service provider	SVM, Gradient boosted decision trees, Naïve Bayes	Accuracy, Precision, Recall, F1	Accuracy:63.83–99.90
J. To examine relationship between socioeconomic and food environment (insecurity)				
Dharmasena et al. (2016)	1. USDA Food Environment Atlas, 2. United States Census Bureau, 3. Center for Disease Control and Prevention, 4. United States Bureau of Labor Statistics, 5. USDA Food and Nutrition Service	Greedy equivalence search (GES)	P-value, T-stat, Std error, Partial value	Std error:0.04–0.44
Nica-Avram et al. (2021)	1. Neighborhood characteristics, 2. Network topology, 3. Behavioral repertoires	RF, AdaBoost, SVM, KNN	Accuracy, Precision, Recall	Accuracy:56.47–75.29
K. To examine the relationship between socioeconomic and health condition				
Mirelman et al. (2016)	1. Health and demographic surveillance system, 2. Survey data 2012	Super learner	Relative risk (RR)	RR: 1.14–1.19
Kolak et al. (2020)	Social Determinants of Health Data	OLS, Spatial autoregressive (SAR)	R2, adj. R2, MSE	R2:0.61, 0.64
L. To predict energy poverty				
Hong and Park (2021)	Household Income and Expenditure survey	Decision tree, ANN, bagging, RF, XGBoost, SVM	Accuracy, Specificity, Precision, Recall, F1	Accuracy:0.89–0.95
Longa et al. (2021)	1. Socioeconomic data at the neighborhood level, 2. A wealth of data at the household level	XGBoost Precision, Recall, F1	F1:75%–91%	
M. To predict poverty using agriculture parameters				
Jiang et al. (2019)	Survey data of poor farmers and basic statistics of villages	Multi-level linear regression model and geographic detector	Q value, P value, and Coefficient	P-value: 0.000–0.878
Chingozha and von Fintel (2019)	Landsat MSS	SVM, OLS	Std.error	−0.91
N. To predict poverty using machine learning				
Verme (2020)	Middle income country	OLS, Logit, RF, Lasso regression	Precision, Accuracy, F2	Accuracy:64.41–69.59
Li et al. (2022)	DHS data	XGBoost, the generalized linear model (GLM)	AUC	0.84–0.91
Sheng (2021)	The poverty-stricken students	RF	Accuracy	75.56–78.61
Kim (2021)	Costa Rican households	RF, Gradient boosting tree	Accuracy, precision, recall, F1	Accuracy: 78.1–79.6

Appendix C

See Table C.1.

Data availability

Data will be made available on request.

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